

SYSTEMATIC SCOPING REVIEW

# Computational Qualitative Analysis with Large Language Models

Methods, human roles, evaluation validity, provenance, confounds, and a  
defensible ACL ARR research agenda

Search cutoff: 24 August 2026 • Target cycle: 12 October 2026 ACL Rolling Review

*Single-reviewer, computationally assisted scoping review; companion evidence workbook supplied*

## Systematic scoping review of LLM and NLP methods for qualitative analysis

**Search cutoff:** 24 August 2026 **Target:** ACL Rolling Review, 12 October 2026 cycle **Review design:** systematic scoping review with critical methodological synthesis **Reporting convention:** *Author claim* reports what a source says. *Reviewer interpretation* is this review's analysis of whether the operation and evidence support that claim. Missing details are recorded as **not reported**; uncertain publication status is recorded as **not verified**.

## Executive synthesis

This review maps 237 unique, full-text-grounded evidence records: 181 core reports with one 12-dimension quality assessment each and 56 adjacent, contextual, or methodological records retained to interpret validity, ethics, and qualitative-method claims. The field has moved rapidly from asking a chat model to suggest codes to constructing hierarchies, applying codebooks at scale, orchestrating role-prompted pipelines, learning persistent instructions from analyst feedback, supporting noncoding theme development, and retaining versioned source lineage. Dai, Xiong, and Ku made codebook revision, rationale, and stopping human decisions ([Dai et al., 2023](#)); LLoM organized concept induction as an analyst-editable loop ([Lam et al., 2024](#)); HICode and Thematic-LM scaled hierarchical or role-prompted induction ([Zhong et al., 2025](#); [Qiao et al., 2025](#)). By the cutoff, NITA kept positionality, counterevidence, theme retention, and the final meta-narrative with human researchers without making coding mandatory; Reflexis logged code evolution and surfaced drift, disagreement, and positionality prompts; and ROCK specified machine-readable source spans, codebooks, relations, and versions ([Nguyen, Trung, and Nguyen, 2026](#); [Ye et al., 2026](#); [Peters et al., 2026](#)).

What is technically possible remains narrower than “automated thematic analysis.” Models can propose manifest codes, apply defined deductive labels, cluster and name concepts, retrieve candidate quotations, draft themes and reports, compare runs, and expose intermediate artifacts for correction. They do not thereby supply researcher positionality, theoretical sensitivity, cultural accountability, or an independent warrant for interpretation. Even apparently strong code application is conditional. Muse reports  $\kappa=.703$  for an ensemble on 12 well-specified codes after per-code threshold tuning, whereas Hill et al. report high percentage agreement but  $\kappa=.34$  in a low-prevalence health task ([Matveyenko et al., 2026](#); [Hill et al., 2026](#)). Dörfel and Ammoneit found 13/14 semantic code mappings above .60 and mean run Jaccard .83, yet coding-unit overlap with the human analysis was .39 and domain-specific distinctions were lost ([Dörfel and Ammoneit, 2026](#)).

The principal weakness is construct validity, not a shortage of metrics. Parkington et al.'s two LLM workflows each missed 190/428 human excerpts; their out-of-box and knowledge-base conditions overlapped only 238/428 and 124/428 excerpts, and 68/251 purported knowledge-base quotations were interpretive summaries rather than quotations ([Parkington et al., 2025](#)). QualiGPT's GPT self-agreement ( $\kappa=.87$ ) exceeded its human-inclusive agreement (about .46), despite human-human  $\kappa=.73$  ([Zhang et al., 2025](#)). IDEAlign's best automated correlations with expert judgments were .618/.683 versus a human oracle's .906/.812; style-only edits inflated embedding similarity by as much as 82% ([Nam et al., 2025](#)). ICR found human reliability of .86–1.00 and LLM reliability of .35–.76 even when cosine and F1 were high ([Pérez et al., 2026](#)). These results do not invalidate similarity or reliability measures; they show why each operation must be validated against its intended qualitative construct.

No-gold evaluation, provenance, reflexivity interfaces, multi-agent consensus, and elementary stopping rules are no longer empty spaces. Aggregate Code Space diagnostics preserve multiple code sets but can reward code flooding and cannot recover concepts omitted by every coder ([Chen et al., 2026](#)). Blind source-symmetric verification shows that lower human-LLM agreement need not imply lower expert preference, but it uses one verifier, one codebook, and code assignment rather than theme interpretation ([Liu et al., 2026](#)). Reflexis and Telescope implement histories, branching, and deliberation; clinical and production pipelines maintain quote or turn identifiers; CAILA, saturation-ratio studies, LOGOS trajectories, and iterative-versus-batch comparisons supply stopping-related signals ([Ye et al., 2026](#); [Lee et al., 2026](#); [Lin et al., 2025](#); [De Paoli and Mathis, 2025](#); [Pi et al., 2025](#)). **Reviewer judgment among the inspected full texts:** what remains open is independent, cross-domain validation of exactness, omission, participant spread, rare and negative cases, disagreement, and auditor usability.

Culturally and clinically situated interpretation reveals bidirectional failure. Japanese clinical work recovered common descriptive themes more readily than low-frequency or culturally specific ones; “fate” was absent ([Sakaguchi et al., 2025](#)). Open models analyzing interviews with Black male firearm-violence survivors produced thousands of labels, averaged about 10% relevance, refused an average 44% of prompts, and erased trauma, sexual-health, racial, and African American English content even when they recovered formal human codes before aggregation ([Zhu et al., 2026](#)). Conversely, LLM-TA overemphasized a rare dramatic event ([Raza et al., 2025](#)). Subbiah et al.'s three-model, five-seed analysis of 154 longitudinal interviews found demographic-conditioned semantic, affective, and thematic shifts; experts reported stereotyping and omission of deaths, violence, abuse, and addiction ([Subbiah et al., 2026](#)). Lukito et al. further showed that topic models missed masking, self-doubt, and strengths and that apparent diagnosis clustering was confounded by interviewer identity ([Lukito et al., 2026](#)). Minority preservation is therefore not a one-directional recall problem: omission, dilution, misattribution, stereotyping, refusal, and sensational overrepresentation all require measurement.

**Reviewer judgment among the inspected full texts:** the strongest October opportunity remains a plurality- and provenance-aware audit of whose voice survives aggregation, but its novelty is the joint validation, not a new ledger or interface. It should retain exact speaker-to-quotation-to-code-to-theme paths; cross participant, interviewer, site, source, and identity perturbations; quantify participant concentration and negative-case survival; and use independent qualitative and domain experts to judge evidence warrants and omissions. VoiceTrace remains Rank 1, WarrantBench Rank 2, and StopTA Rank 3, conditional on immediate access to authorized data and evaluators.

## 1. Introduction and motivation

Qualitative analysis turns situated language into claims about experience, practice, meaning, and social structure. Its labor is not merely labeling: analysts read and reread a corpus, memo, compare cases, revise concepts, inspect contradiction, locate counterevidence, and write an interpretation whose warrant should remain visible in the data. Fluent codes and plausible themes are easy for an LLM to generate; establishing methodological congruence, evidential sufficiency, representativeness, reflexivity, and safety is much harder.

The reviewed literature spans different tasks and epistemologies. Some papers predict a short label for a presegmented quotation; others apply a fixed codebook, induce a hierarchy, create themes, build an interactive workbench, support noncoding narrative synthesis, or evaluate competing code spaces. Treating all of these as one “thematic analysis” benchmark erases the claim being tested. Parfenova et al. evaluate one-code-per-segment generation rather than axial or theme construction ([Parfenova et al., 2025](#)); QuaLLM combines extraction, classification, aggregation, and prevalence estimation ([Rao et al., 2025](#)); NITA treats coding as optional and keeps interpretive synthesis with researchers ([Nguyen, Trung, and Nguyen, 2026](#)). A fixed-label F1 or  $\kappa$  claim can be sound without licensing a claim of equivalent reflexive, grounded-theory, or clinical interpretation.

This review therefore asks both what systems do and what their evidence licenses us to conclude. It covers 181 core computational or empirical reports and 56 adjacent, contextual, and methodological records through 24 August 2026. Preprints, accepted manuscripts, posters, protocols, and position papers are retained where they define the frontier, but are labeled separately from verified peer-reviewed work. Publication families are counted once at the preferred verified version, with event-year, online-first, preprint, and formal-proceedings histories preserved where they differ.

## 2. Review questions

The review addresses eight questions.

1. **RQ1—tasks and methods:** Which parts of familiarization, quotation selection, open or deductive coding, codebook induction/refinement, axial or hierarchical coding, noncoding theme development, subtheme/theme creation, evidence retrieval, memo/schema construction, conceptual modeling, reporting, confidence/abstention, stopping, and quality assurance are automated or assisted?
2. **RQ2—methodological grounding:** Does each study claim and operationalize reflexive TA, codebook TA, coding-reliability TA, grounded theory, qualitative content analysis, Framework analysis, rapid analysis, noncoding qualitative synthesis, generic coding, or topic discovery? Where does the implemented operation narrow or conflict with the label?
3. **RQ3—architecture:** Which model versions, prompt/adaptation strategies, agents or ensembles, retrieval and graph components, chunking/context strategies, stochastic controls, confidence/diversity routes, evidence links, stopping rules, and intervention points form the workflow?
4. **RQ4—human-AI relationship:** Is the human absent after setup, a question/codebook supplier, a reference creator, a post-hoc checker, an iterative collaborator, an adjudicator, a domain or stakeholder validator, or the primary analyst with authority to reject, backtrack, memo, and stop?
5. **RQ5—evaluation:** What construct does each automatic or human operation purport to measure; is it adequately operationalized, independently validated, compatible with plurality, calibrated, source-grounded, and statistically supported; and does workload include verification?
6. **RQ6—validity threats:** Could results arise from participant, source, site, diagnosis, platform, language, time, boilerplate, interviewer speech, length/position, overlap/leakage, contamination, lexical style, same-family judging, anchoring, majority dominance, overconfidence, or omitted negative cases?
7. **RQ7—ethics:** Are consent, ethics review, privacy, deployment, retention/training terms, bias, stakeholder involvement, credential/release security, accountability, and reproducibility restrictions reported?
8. **RQ8—opportunity:** Which gaps remain after version-family deduplication and full-text inspection of the 2025–2026 wave, and which can support a defensible October ARR paper?

### 3. Search and screening methodology

#### 3.1 Scope, protocol, and eligibility

Eligible primary work presents, evaluates, or substantively analyzes computational support for qualitative coding or interpretation; proposes evaluation directly applicable to codes or themes; studies human-AI qualitative analysis; or isolates a validity mechanism that changes such evaluation. Methodological sources were retained to assess congruence. Pure topic modeling, summarization, sentiment analysis, or fixed-label prediction without a directly relevant qualitative contribution; studies that merely use manual TA to study another technology; commercial pages; and unverifiable records were excluded from the core. Adjacent records were retained when they supplied directly relevant evidence about constructs, confounding, privacy, or workflow design.

The search began with ten specified seeds and five supplied works and added backward references, focused forward-title/citation queries, same-author and publication-family searches, and searches for systems, datasets, and metrics cited by eligible studies. A preprint and proceedings or journal paper were treated as one family unless their methods materially differed. DeTAILS remains a material exception: the seven-page CUI paper and its expanded preprint have different evaluative content ([official CUI paper](#); [expanded preprint](#)). Citation chaining was focused and source coverage differed; no exhaustive citation-saturation claim is made.

#### 3.2 Databases, dates, and identification

All auditable searches were run on 24 August 2026 with a 1 January 2020–24 August 2026 date window where supported. Within-source deduplication produced 295 PubMed, 688 Crossref, 13 ACL Anthology, 120 OpenReview, and 440 arXiv records. Their union contained 1,556 records; exact DOI and normalized-title deduplication removed 196, leaving 1,360 unique database records.

| Source        | Search implementation and filters                                        | Returned/retained for union   | Access caveat                                               |
|---------------|--------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------|
| PubMed        | Seven Title/Abstract Boolean queries; 2020-01-01 to 2026-08-24           | 312 returns; 295 unique PMIDs | Complete machine export                                     |
| ACL Anthology | Official BibTeX snapshot; title-token seed/task queries; 2020–2026       | 20 returns; 13 unique         | Title-only retrieval supplemented by full text and chaining |
| arXiv         | Seven task queries plus identity/shortcut query; cutoff enforced locally | 440 unique                    | Identity query capped at 200 of 241 reported                |
| OpenReview    | Nine exact-phrase searches with client-side relevance filter             | 120 candidates                | Retained count was algorithmic triage, not inclusion        |

| Source                            | Search implementation and filters                                   | Returned/retained for union | Access caveat                                                    |
|-----------------------------------|---------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------|
| Crossref                          | Seven relevance-ranked title searches; first 100/query; date filter | 700 returns; 688 unique     | Fuzzy multi-million totals were not treated as screenable counts |
| ACM DL / IEEE Xplore              | Targeted title, venue, DOI, author, and chain queries               | Count not reported          | No stable auditable bulk total was exposed                       |
| Google Scholar / Semantic Scholar | Targeted seed-title and citing-paper discovery                      | Count not reported          | Rate/access limits prevented auditable bulk export               |
| Web of Science / Scopus           | Intended in protocol                                                | Not accessible              | No institutional interface; no coverage claim                    |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The query families combined LLM/generative-AI terms with thematic analysis, qualitative/open/inductive/deductive coding, qualitative content analysis, collaboration, agents, codebooks, hierarchy/schema/concept induction, evaluation, provenance, auditability, identity/shortcut/confounding, and clinical interviews. Appendix 19.1 preserves the exact concept strings and source-specific exports.

### 3.3 Terminal screening and PRISMA-style reconciliation

Initial deterministic triage divided the 1,360-record union into 122 high-priority, 612 uncertain, and 626 low-priority records. Those were prioritization labels, not eligibility decisions. The completed second passes gave every database record a terminal title/abstract disposition: 1,116 were excluded and 244 advanced to full-text resolution. Of the 244, 44 were not retrieved; 200 were assessed or resolved, yielding 174 retained reports, 17 full-text exclusions, and nine publication/version-family resolutions.

Focused forward chaining generated 59 route records, but 21 already occurred in the database union. The identity-based outside-union flow was therefore 38 screened, four title/abstract exclusions, 34 sought, eight not retrieved, 26 assessed, one full-text exclusion, and 25 retained reports (16 core and nine adjacent). Three backward-chain reports were sought; one was not retrieved and two were assessed and retained. The secondary route-event ledger also records 20 prespecified, supplied, or targeted retrieval events; after identity reconciliation, the broader “other” stratum contains 36 unique prespecified, supplied, targeted, or methodological-foundation reports.

The primary identity-deduplicated full-text flow is 317 unique reports sought, 53 not retrieved, and 264 assessed or resolved; the latter comprise 237 retained reports, 18 full-text exclusions, and nine publication/version-family resolutions. Equivalently,  $317 = 53 + 264$  and  $264 = 237 + 18 + 9$ . The 237 retained identities comprise 181 core and 56 noncore reports. The route-event table below is retained as a secondary audit because it records how often retrieval pathways were exercised, including overlap.

The fully inspected [Kempny et al. secondary review](#) is deliberately separated from the evidence identity ledger: TA-0428 is an assessed **evidence-map exclusion** in `review_flow.json`, while Section 18.3 retains it as an additional contextual citation outside the 237-row master. Its reporting-practice statistics are therefore full-text-grounded, but it is not counted or quality-scored as retained primary/adjacent evidence.

| Full-text accounting route                    | Sought     | Not retrieved | Assessed/resolved | Retained   | Full-text excluded | Family resolution |
|-----------------------------------------------|------------|---------------|-------------------|------------|--------------------|-------------------|
| Database/register union                       | 244        | 44            | 200               | 174        | 17                 | 9                 |
| Focused forward route events                  | 54         | 10            | 44                | 41         | 1                  | 2                 |
| Backward chaining                             | 3          | 1             | 2                 | 2          | 0                  | 0                 |
| Other prespecified/supplied/targeted evidence | 20         | 0             | 20                | 20         | 0                  | 0                 |
| <b>All documented route events</b>            | <b>321</b> | <b>55</b>     | <b>266</b>        | <b>237</b> | <b>18</b>          | <b>11</b>         |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The route-event table is auditable but is not an identity-addition table: its focused-forward row includes 21 database-union overlaps. Its arithmetic is exact: 321 sought route events = 55 not retrieved + 266 assessed/resolved, and 266 = 237 retained canonical evidence records + 18 full-text exclusions + 11 family resolutions. The identity-based forward reconciliation above prevents those 21 overlaps from being added to the 1,360-record database denominator. All nonretrievals remain explicit; no abstract-only report supplies a method or numerical result in the synthesis.

Screening and extraction were performed by one reviewer with computational assistance. There was no independent duplicate screen or consensus adjudication. The terminal flow removes the old unresolved-triage limitation, but single-reviewer decisions, 53 unretrieved unique reports (55 nonretrieval route events), focused rather than exhaustive chaining, inaccessible Web of Science/Scopus, and nonauditable bulk Scholar access still limit completeness. The row-level flow records parent seed/source, title/abstract reason, retrieval status, full-text decision, exclusion reason, and family resolution.

### 3.4 Extraction and quality assessment

The canonical master contains 237 unique evidence records with 52 fields covering bibliography, method claim and operation, architecture, data, human role, evaluation, results, ethics, resources, limitations, dependencies, and ARR relevance. Missing information is recorded as **not reported**. Every one of the 181 core records has exactly one quality assessment on twelve 0/1/2 dimensions: method clarity, dataset appropriateness, workflow transparency, baseline independence, human evaluation, automatic-metric validity, plurality, evidence traceability, stability, responsible reporting, resource availability, and claim proportionality. The 56 adjacent/contextual/methodological records are unscored; nine provisional adjacent scores were removed during consolidation. Totals are navigation aids, not a study leaderboard.

## 4. Conceptual foundations of thematic analysis

“Thematic analysis” is a family, not a single algorithm. Coding-reliability TA usually applies a comparatively structured frame and can make agreement and adjudication appropriate. Codebook TA uses a structured coding framework while retaining interpretive judgment. Reflexive TA treats researcher subjectivity and iterative engagement as analytic resources; themes are constructed patterns of shared meaning, not labels waiting to be recovered. Braun and Clarke explicitly warn against one-size-fits-all quality criteria and against importing consensus coding or saturation into reflexive TA as universal tests ([Braun and Clarke, 2021a](#); [Braun and Clarke, 2021b](#)).

Grounded theory requires more than “open” and “axial” labels: constant comparison, memoing, theoretical sampling, category integration, and theory development are central, with axial coding belonging to some schools rather than all ([Corbin and Strauss, 1990](#)). Qualitative content analysis can be conventional, directed, or summative, which changes whether categories arise from the data, prior theory, or counted content plus contextual interpretation ([Hsieh and Shannon, 2005](#)). Framework analysis preserves a case-by-code matrix and is well suited to transparent team comparison ([Gale et al., 2013](#)). Rapid qualitative analysis uses focused templates and matrices for time-sensitive questions; its quality depends on purpose, expertise, omissions, and verification, not speed alone ([PARROQA, 2024](#)).

These distinctions matter computationally. A model that predicts a short code for each sentence can be evaluated as labeling. A model that constructs latent themes in RTA cannot be validated solely by matching one analyst's wording. A grounded-theory system should expose comparisons, memos, category relations, and sampling decisions rather than merely cluster open codes. A Framework tool should retain case context and matrix provenance. The common pipeline—segment text, generate labels, embed or LLM-merge them, compare the output with a consensus set—is usually most congruent with coding-reliability or codebook TA, even when a paper invokes reflexive language.

This is not a purely terminological objection. McDonald, Schoenebeck, and Forte found substantial method-label ambiguity in 251 CSCW papers: only 20 of 85 papers reporting open coding described grounded theory, and fewer than half of the 161 papers coding to identify themes named an analysis method ([McDonald et al., 2019](#)). **Reviewer interpretation:** computational papers inherit this ambiguity and then add a measurement error: they often translate “similar to a reference codebook” into “valid qualitative interpretation.” The appropriate remedy is an explicit analytic contract stating the research question, unit, tradition, claim level, human role, expected product, and success criteria before running the model.

## 4.1 Claimed versus operationalized methodology in the core corpus

The claimed-versus-operationalized audit was recomputed over all 181 core records in the frozen master. Both fields contain substantive information for 169 records; ten report neither clearly enough to code; [ashwin\\_2025\\_serious\\_bias](#) contains a claim without an adequately specified operation, and [bai\\_2026\\_three\\_datasets](#) contains an operational description without a clear tradition claim. These are reporting-coverage counts, not congruence scores. A nonempty “thematic analysis” field does not show that the implementation follows the named tradition.

A strict lower-bound reviewer audit flagged 34 of 181 core reports as containing a material claim–operation mismatch or narrowing. The rule did **not** flag sparse reporting or weak evaluation alone, and it did not flag fixed-label classification when a paper explicitly claimed deductive coding. Each flagged record was assigned one dominant category to avoid double counting. The 34/181 figure is therefore a reproducible reviewer judgment under this rubric, not an estimate of author error or a timeless field prevalence.

| Dominant claim–operation discrepancy                                                                   | Count | Representative operation and inspected examples                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Named interpretive/reflexive tradition replaced by codebook, reliability, or automated-schema practice | 8     | Hill evaluates a reflexive-TA claim through an adjudicated codebook and inter-rater reliability; Parkington uses a hierarchical codebook and excerpt-overlap comparison; the automated batching study invokes Braun–Clarke phases but evaluates theme-set Jaccard ( <a href="#">Hill et al., 2026</a> ; <a href="#">Parkington et al., 2025</a> ; <a href="#">Logan et al., 2026</a> ). |
| Inductive or no-predetermined-frame claim becomes fixed, supervised, or reference-conditioned          | 5     | SFT-TA trains inside a twelve-theme reference universe; Hairston predicts twelve inherited themes; Shah calls zero-shot fixed multiclass labeling “inductive” ( <a href="#">SFT-TA</a> ; <a href="#">Hairston et al., 2025</a> ; <a href="#">Shah et al., 2025</a> ).                                                                                                                   |
| Grounded-theory or omnibus method-family claim reduced to schema grouping or generic templates         | 5     | Rasheed gives five traditions the same summarize/code/theme/report roles; LOGOS and Neo-Grounded Theory optimize schemas without demonstrating theoretical sampling or validated theory development ( <a href="#">Rasheed et al., 2024</a> ; <a href="#">Pi et al., 2025</a> ; <a href="#">Wen et al., 2025</a> ).                                                                      |

| Dominant claim-operation discrepancy                                                                                                                    | Count | Representative operation and inspected examples                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full qualitative/thematic/content-analysis claim narrowed to classification, extraction, clustering, factor/topic reduction, or another proxy component | 15    | Mackenzie extracts legal fields; Ghali forces a three-cluster embedding partition then names it; Beltrán generates short codes, reduces a manifold, clusters, and names the clusters ( <a href="#">Mackenzie et al., 2025</a> ; <a href="#">Ghali et al., 2025</a> ; <a href="#">Beltrán et al., SEET 2025 event/2026 proceedings</a> ). |
| Explicitly claimed mechanism absent from the evaluated implementation                                                                                   | 1     | Reinforcement learning is central to Auto-TA's title and framing, but RLHF/PPO is not implemented in the reported experiments ( <a href="#">Auto-TA preprint/ACL-SRW poster</a> ).                                                                                                                                                       |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The audit also identifies comparatively congruent patterns. Bounded codebook and qualitative-content-analysis studies can appropriately target code assignment, agreement, or downstream estimates if the units and labels are independently justified. NITA is congruent with its noncoding, researcher-primary claim because positionality, counterevidence, theme retention, and the final synthesis remain human responsibilities ([Nguyen, Trung, and Nguyen, 2026](#)). Reflexis operationalizes several reflexive commitments through optional prompts, preserved disagreement, immutable code histories, and analyst control, although its short, synthetic-corpus study does not establish improved interpretation ([Ye et al., 2026](#)). Ronaghi et al. likewise match their task-specific claim to a human-final workflow: models organize researcher-authored summaries in one task and retrieve, verify, and sort quote-grounded evidence across 167 interviews in another; humans retain contextual interpretation and intervention decisions ([Ronaghi et al., 2026 preprint](#)).

**RQ2 answer.** The corpus includes reflexive, codebook, and coding-reliability TA; grounded-theory claims; qualitative content, Framework, and rapid-analysis variants; generic open or deductive coding; topic/concept discovery; and deliberate noncoding synthesis. Their success criteria are not interchangeable. Fixed-label F1 or  $\kappa$  supports a bounded assignment claim; cluster coherence supports organization; reference similarity supports proximity to one reading; none alone establishes reflexive interpretation, grounded-theory development, or clinically warranted themes. The complete 181-row extraction preserves each author's label beside the reviewer's operational description so readers can audit this distinction paper by paper.

## 5. Taxonomy of computational approaches

### 5.1 Historical trajectory

Pre-LLM systems used co-occurrence, latent semantic analysis, topic models, embeddings, clustering, and supervised classifiers to organize text. The conceptual break introduced by instruction-tuned LLMs is not that clustering disappeared; rather, natural-language generation could draft codes, definitions, evidence rationales, and hierarchical relations while accepting research questions and codebooks in ordinary language.

The resulting literature is better represented by ten nonexclusive operational families. A single system can occupy several families; the list is a task map, not ten mutually exclusive paper bins.

9. **Fixed-label or codebook assignment.** Dai et al. co-construct and then apply a codebook; Muse, QuaLLM, Turner et al., Than et al., and many applied studies assign defined codes to segments ([Dai et al., 2023](#); [Matveyenko et al., 2026](#); [Rao et al., 2025](#); [Turner et al., 2026](#); [Than et al., 2025](#)). This is the regime in which F1, accuracy, or  $\kappa$  has the clearest target, provided prevalence, units, and independent labels are reported.
10. **Open-code suggestion and collaborative code development.** Qualitative Code Suggestion, CollabCoder, Dai, DeTAILS, LLMCode, MindCoder, and ROSAnnotator expose candidate codes for human acceptance, editing, rejection, or comparison ([Spinoso et al., 2023](#); [Gao et al., 2024](#); [Sharma and Wallace, 2025](#); [Oksanen et al., 2025](#); [Gao et al., 2025](#); [Zhang et al., 2025](#)). Suggestion uptake is not independent validity because seeing a candidate can anchor the analyst.
11. **Hierarchy, schema, axial, and theory-like refinement.** HICode, LOGOS, Neo-Grounded Theory, and newer open-axial-selective or taxonomy systems merge codes, relate categories, or optimize schemas ([Zhong et al., 2025](#); [Pi et al., 2025](#); [Wen et al., 2025](#); [Liu et al., 2026](#)). These operations implement structure; without theoretical sampling, constant comparison, memo integration, and evaluated theory development, they do not establish grounded-theory automation.
12. **Code-first end-to-end theme pipelines.** LLM-TA, Auto-TA, SFT-TA, TAMA, and clinical refinement systems sequence quotation/code generation, consolidation, themes, evaluation, and report-like outputs ([Raza et al., 2025](#); [Yi et al., 2025](#), [Auto-TA preprint/poster](#); [Yi et al., 2025](#), [SFT-TA preprint/non-archival poster](#); [Xu et al., 2026](#); [Yi et al., 2026](#)). An end-to-end artifact chain is not evidence of end-to-end interpretive validity.
13. **Noncoding narrative and theme development.** NITA explicitly makes coding optional and keeps positionality, counterevidence, theme retention, and the meta-narrative with researchers

([Nguyen, Trung, and Nguyen, 2026](#)). This family matters conceptually because “theme generation” need not be implemented as clustering increasingly abstract labels.

14. **Interactive workbenches and reflexivity support.** [LLooM](#), [DeTAILS](#), [Muse](#), [CentaurTA](#), [Co-Refine](#), [Reflexis](#), and [Teleoscope](#) provide editable concepts, persistent feedback, drift/disagreement prompts, or branchable analytic graphs. Reflexis shows that deliberation and positionality interfaces exist; the open question is whether they improve independently judged interpretation over time.
15. **Autonomous, role-prompted, and deliberative pipelines.** [Thematic-LM](#), [TA-MAS](#), [Auto-TA](#), [TAMA](#), and related systems assign coder, aggregator, reviewer, or critic roles. “Multi-agent” can mean one model repeated under role prompts, same-model deliberation, or genuinely different model families. Borchers et al. found only 5 of 432 reported contrasts favoring multi-agent consensus and 65 favoring single agents, so consensus cannot be assumed to provide independent triangulation ([Borchers et al., 2026](#)).
16. **Adaptation, fine-tuning, and steering.** Parfenova et al. use LoRA for short-code generation; Ganesh et al. show task-specific fine-tuning can outperform in-context learning; SFT-TA trains on paraphrases derived from ten of twelve reference themes; and the AIED confidence study tests activation steering ([Parfenova et al., 2025](#); [Ganesh et al., 2024](#); [Yi et al., 2025](#); [Li et al., AIED 2026 manuscript/formal 2027 proceedings](#)). Leakage-safe held-out concepts, domains, model families, and dates are necessary before calling adaptation general.
17. **Retrieval, evidence, provenance, and version control.** [Clinical pipelines](#), [Nyaaba](#), [MindCoder](#), [ROSAAnnotator](#), [Reflexis](#), [Teleoscope](#), and [ROCK](#) connect claims to quotations or source spans and increasingly retain revisions and branches. Traceability is an enabling condition; exactness, sufficiency, participant spread, counterevidence, and independent-auditor usability remain separate empirical questions.
18. **Evaluation, confidence, disagreement, and stopping.** Aggregate Code Space, source-symmetric verification, IDEAlign, ICR, LLM-judge calibration, rationale-based triage, repeated-run comparisons, saturation ratios, and refinement trajectories make assurance a first-class computational task ([Chen et al., 2026](#); [Liu et al., 2026](#); [Nam et al., 2025](#); [Pérez et al., 2026](#); [Han et al., 2026](#); [Tajik et al., 2026](#); [De Paoli and Mathis, 2025](#)). Severe confidence miscalibration and anchoring show that a numeric or verbal confidence score is not itself a trustworthy routing signal.

## 5.2 Comparative architecture table

| System           | Operational task                                                                | Architecture/adaptation                                                  | Actual human intervention and lineage                                                       | Central caveat/status                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dai 2023         | Co-develop codebook; assign codes                                               | GPT-3.5, few-shot examples, temperature 0                                | Analysts edit codes, give rationales, and decide when to stop; assignments link to excerpts | The LLM helps construct the reference it later matches ( <a href="#">official paper</a> ).                                                                                        |
| Lloom 2024       | Induce and score concepts                                                       | LLM operators plus embeddings/HDBSCAN                                    | Analysts edit, merge, split, and rescore concepts                                           | Two short expert sessions; concept induction, not a formal TA validation ( <a href="#">official paper</a> ).                                                                      |
| HiCode 2025      | Generate codes and a hierarchy                                                  | Segment labels plus repeated 100-label clustering                        | Goal/context supplied; no evaluated analyst intervention in main generation                 | Embedding matches and human mappings differ; original annotators' $\alpha=.31$ ( <a href="#">official paper</a> ).                                                                |
| Thematic-LM 2025 | Codes, codebook, themes                                                         | AutoGen coder/aggregator/reviewer; retrieval and compression             | No independent analyst in the evaluated main loop                                           | Same-family evaluation and ROUGE-based trustworthiness proxies ( <a href="#">official paper</a> ).                                                                                |
| DeTAILS 2025     | Editable codebook, global coding, clustering, themes                            | RAG-assisted interactive workflow                                        | Analysts review text and edit proposed codes/themes                                         | Helpful uptake can also be anchoring; CUI paper and expanded preprint contain different evaluation detail ( <a href="#">official paper</a> ; <a href="#">expanded preprint</a> ). |
| NITA 2026        | Noncoding familiarization, pattern development, counterevidence, meta-narrative | Staged generative prompts and researcher deliberation                    | Researchers own positionality, retention/rejection, counterevidence, and final narrative    | Researcher-primary workflow on one dependency family; not autonomous coding ( <a href="#">journal VOR</a> ).                                                                      |
| CentaurTA 2026   | Open codes and themes with persistent feedback                                  | Actor/Critic plus prompt-principle optimizer                             | Six users provide feedback converted into reusable instructions                             | Small study; judge/rubric dependence; performance can decline after repeated updates ( <a href="#">official paper</a> ).                                                          |
| Reflexis 2026    | Reflexive code comparison and deliberation                                      | Drift/disagreement detection, positionality prompts, immutable histories | Twelve researchers in six pairs may ignore prompts and retain disagreement/control          | No baseline; synthetic corpus; author-rated prompts; short sessions ( <a href="#">official paper</a> ).                                                                           |

| System                | Operational task                                                      | Architecture/adaptation                                                            | Actual human intervention and lineage                                                                      | Central caveat/status                                                                                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ronaghi 2026          | Organize site summaries; deductively retrieve and synthesize evidence | Human task decomposition, embeddings/RAG, quote verification and relevance sorting | Researchers build the manual exemplar, inspect sources, interpret context, and make intervention decisions | Preprint; RAG fragmentation produced tangential retrieval and overgeneralization ( <a href="#">official preprint</a> ).                                                                                                                                                           |
| ROCK 2026             | Standardize source-to-code-to-theme artifacts                         | Machine-readable spans, codebooks, relations, and versions                         | Defines interoperable lineage rather than replacing analyst judgment                                       | Preprint standard; adoption, privacy, and audit-outcome gains are not yet validated ( <a href="#">official preprint</a> ).                                                                                                                                                        |
| Auto-TA / SFT-TA 2025 | Automated code/theme roles; reference-conditioned adaptation          | Sequential agents; SFT-TA fine-tunes on paraphrased themes                         | Humans create the source/reference outside the tested main loop                                            | Auto-TA is arXiv v2 plus <a href="#">ACL-SRW 2025 Poster</a> , not a verified archival ACL paper; SFT-TA is arXiv v1 plus <a href="#">ML4H 2025 Reject from Proceedings as Findings Poster</a> , not an archival PMLR paper ( <a href="#">Auto-TA</a> ; <a href="#">SFT-TA</a> ). |
| Li confidence study   | Nine-class educational-dialogue coding and abstention signals         | GPT/Gemini/Llama prompts, confidence anchors, activation steering                  | Human labels evaluate fixed-code outputs; no live analyst workflow                                         | Confidence is severely miscalibrated; AIED 2026 manuscript, formal Springer citation dated 2027 ( <a href="#">official chapter</a> ).                                                                                                                                             |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

**RQ1 answer.** Implemented support now covers familiarization proxies; quotation selection; open and deductive coding; codebook induction, reduction, and refinement; axial/hierarchical grouping; subtheme and theme generation, naming, definition, and refinement; noncoding pattern and narrative development; evidence retrieval and quotation verification; memo/schema construction; report generation; provenance/version control; disagreement and confidence triage; repeated-run comparison; stopping signals; and automatic or human quality assurance. Familiarization is usually summarized or retrieved rather than evaluated as sustained engagement. Conceptual models and theory-like schemas are generated, but inspected evidence does not yet establish full grounded-theory development. Simple no-new-theme, saturation-ratio, fixed-round, iterative-versus-batch, and analyst-confidence stopping signals exist; their held-out relationship to supported minority/negative-case recall and verification burden remains unvalidated.

**RQ3 answer.** Architectures span proprietary and open-weight models, zero/few-shot prompting, LoRA or supervised fine-tuning, retrieval, clustering and graphs, same-model role agents, cross-model ensembles, repeated runs, self/cross critique, confidence routing, and human-editable provenance. Exact snapshots, chunk order, context strategy, seeds, retries, and human edits remain inconsistently reported. A role label does not confer independent expertise, a temperature of zero does not guarantee determinism, and a source link does not establish a warranted or representative interpretation.

## 6. Human-AI collaboration models

“Human in the loop,” “collaboration,” and “automation” describe very different interventions. This review classifies the implemented decision point and authority, not the paper's preferred label.

| Implemented human relationship           | Representative systems                                                     | What the person actually does                                                                                                  | Reviewer assessment                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No intervention after setup              | HICode, Thematic-LM, Auto-TA, automated provenance/refinement              | Supplies a goal, background, or starting corpus; an LLM “reviewer” may revise another LLM                                      | Automated for the tested run, not collaborative interpretation ( <a href="#">HICode</a> ; <a href="#">Thematic-LM</a> ; <a href="#">Auto-TA</a> ; <a href="#">provenance/refinement</a> ).                                                                                                                                                                                                             |
| Question, examples, or codebook supplied | QuaLLM, Turner, Muse apply-code, fixed-label studies                       | Defines the question, categories, decision rules, and sometimes examples                                                       | Appropriate bounded assistance if the codebook and evaluation labels are independent; not evidence of inductive theme discovery ( <a href="#">QuaLLM</a> ; <a href="#">Turner</a> ; <a href="#">Muse</a> ).                                                                                                                                                                                            |
| Reference creator or post-hoc checker    | Mackenzie, HICode subset, TAMA, Misra                                      | Creates the human reference, judges outputs after generation, or stops refinement                                              | Valuable expertise, but “in the loop” can overstate intervention and reuse of one's own themes reduces independence ( <a href="#">Mackenzie</a> ; <a href="#">HICode</a> ; <a href="#">TAMA</a> ; <a href="#">Misra et al., 2026</a> ).                                                                                                                                                                |
| Reviews or edits proposals               | Dai, LLoOM, DeTAILS, HHITL-TA, LLMCode, MindCoder, ROSAnnotator, CentaurTA | Accepts, rejects, merges, revises, explains, backtracks, or stops                                                              | Substantive control is stronger than post-hoc checking, but exposure to machine suggestions can anchor the human; pre/post artifacts and rejected alternatives should be logged ( <a href="#">Dai</a> ; <a href="#">LLoOM</a> ; <a href="#">DeTAILS</a> ; <a href="#">HHITL-TA</a> ; <a href="#">LLMCode</a> ; <a href="#">MindCoder</a> ; <a href="#">ROSAnnotator</a> ; <a href="#">CentaurTA</a> ). |
| Pair or team deliberation                | Reflexis, Nyaaba                                                           | Compares disagreements, receives optional reflexivity prompts, writes memos, revises interpretations, and can ignore AI advice | Reflexis preserves disagreement and control; Nyaaba documents staged Ghanaian researcher chats, revisions, and memos. Neither by itself proves improved downstream validity ( <a href="#">Reflexis</a> ; <a href="#">Nyaaba</a> ).                                                                                                                                                                     |

| Implemented human relationship           | Representative systems                                                        | What the person actually does                                                                                                                             | Reviewer assessment                                                                                                                                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Independent adjudicator or domain expert | Hill; blind source-symmetric K-12 comparison; specialist clinical comparisons | Judges evidence or competing outputs while blinded where possible                                                                                         | Stronger when reference creation is separated, expertise and sampling are explicit, and disagreement/uncertainty are retained ( <a href="#">Hill</a> ; <a href="#">Agreement Is Not Quality</a> ; <a href="#">specialist/lay clinical comparison</a> ). |
| Participant or stakeholder validator     | Rare; some source analyses were previously member-checked                     | Assesses whether an AI-assisted interpretation fits lived experience or consequences                                                                      | An inherited member check validates the source analysis, not a later model output ( <a href="#">Zhu et al.</a> ). AI-specific participant/stakeholder validation remains uncommon among inspected full texts.                                           |
| Human remains primary analyst            | NITA, Vikan, Ronaghi                                                          | Owns positionality, counterevidence, interpretive synthesis, and final decisions; the model organizes, retrieves, suggests, or verifies bounded artifacts | Most congruent with reflexive or high-stakes interpretation, but may save no time and still requires audits for anchoring and omission ( <a href="#">NITA</a> ; <a href="#">Vikan</a> ; <a href="#">Ronaghi</a> ).                                      |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The expanded DeTAILS preprint reports high suggestion-to-revision F1. That can indicate useful alignment, but without a no-suggestion arm it cannot distinguish assistance from researchers retaining the machine's framing ([Sharma et al., 2025 expanded preprint](#)). CentaurTA makes feedback persistent, yet scores can decline after repeated updates, so “self-improving” is a mechanism with local evidence, not monotonic general learning ([Wang et al., 2026](#)). Reflexis reports that 11/12 researchers valued its disagreement focus and 12/12 felt they retained control, while its two author-raters judged all seven triggered drift alerts and 18 prompts relevant; the lack of a baseline, synthetic corpus, and nonindependent ratings limits outcome claims ([Ye et al., 2026](#)).

The K-12 instrument pipeline provides a procedural counterexample to token oversight. Researchers read raw data beside model output, memoed misleading or generic proposals, retained every merge/category decision, and then three trained coders applied the resulting 67-item hierarchy to a message- and conversation-disjoint 2,560-message sample and added five missed codes ([Liu et al., 2026](#)). Its broader corpus contains 45,000 messages; the inductive phases used 21,147 overlapping three-message windows. These denominators describe different analytic stages and must not be interchanged.

**RQ4 answer.** Human roles range from absent after setup, through question/codebook supply and post-hoc reference checking, to iterative editing, adjudication, pair deliberation, domain validation, and researcher-primary interpretation. Claims of collaboration are accurate only when a person can intervene in the live analysis and retains consequential authority. The most defensible workflows expose machine suggestions and evidence, log edits/rejections/rationales, preserve disagreement, permit abstention and backtracking, and separate reference creators from evaluators where feasible.

## 7. Datasets and application domains

The empirical base is broader by document count than by qualitative setting. Large corpora include Reddit, forum posts, news, litigation/OCR segments, academic abstracts, and survey responses; health work adds interviews and focus groups that are smaller but methodologically consequential. HICode spans four corpora from 369 documents to 163,173 OCR segments ([Zhong et al., 2025](#)). QuaLLM processes 65,377 Reddit submissions and 1,392,776 comments, retaining 58,728 concerns after an 11% API-failure rate ([Rao et al., 2025](#)). Muse releases an eleven-dataset evaluation collection spanning interviews, news, academic/medical abstracts, social media, legal transcripts, and discussion data ([Matveyenko et al., 2026](#)). These studies establish computational scale, but arXiv category labels or Reddit topics are not a proxy for latent interpretation of a sensitive interview.

| Dataset/domain cluster      | Scale and analytic use                                                                                                                                                      | Evidence contribution                                         | Main limitation                                                                                                                                                                                        |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dai survey responses        | Music: 35 analyzed answers; password manager: 100 development plus small seen/unseen tests                                                                                  | Early iterative human-LLM codebook/coding                     | Very small evaluation subsets and possible public-data contamination ( <a href="#">source</a> )                                                                                                        |
| Parfenova quote-code corpus | 600 social-science pairs + 400 SemEval reviews; 900/100 train/test                                                                                                          | Open-weight/LoRA short-code benchmark                         | Quote segmentation and one-label target remove corpus-level interpretation ( <a href="#">source</a> )                                                                                                  |
| HICode corpora              | Frame, Astro, Values, OIDA; up to 163,173 segments                                                                                                                          | Scale and hierarchical schema induction                       | Reference hierarchies and embedding thresholds remain domain/task dependent ( <a href="#">source</a> )                                                                                                 |
| Online forums/social media  | QuaLLM, Thematic-LM, DeTAILS, Dreaddit                                                                                                                                      | Scale, prevalence, interactive tool testing                   | Public availability does not remove contextual-integrity or consent concerns ( <a href="#">QuaLLM</a> ; <a href="#">Thematic-LM</a> ; <a href="#">DeTAILS</a> )                                        |
| AAOCA clinical focus groups | Nine groups, 42 parents, roughly 11,000 words/group, 12 reference themes                                                                                                    | End-to-end coding/theme/refinement benchmark                  | Reused by <a href="#">LLM-TA</a> , <a href="#">Auto-TA</a> , <a href="#">SFT-TA</a> , <a href="#">TAMA</a> , and <a href="#">later provenance work</a> ; one dependency cluster, not five replications |
| K-12 educator-AI messages   | 45,000 messages total, including a message- and conversation-disjoint 2,560-message human-validation sample; inductive phases used 21,147 overlapping three-message windows | Auditable open/axial/selective pipeline and human-added codes | One platform/domain; commercial model proposals are not strictly reproducible ( <a href="#">source</a> )                                                                                               |

| Dataset/domain cluster                    | Scale and analytic use                                                        | Evidence contribution                                                                         | Main limitation                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GATOS synthetic corpora                   | 854 teammate-feedback, 823 ethics-culture, and 1,110 return-to-work responses | Controlled recovery of planted themes using open-weight Mistral-22B-2409 workflow             | Synthetic generator distribution and short responses can make recovery easier than live interpretation ( <a href="#">source</a> )                                                 |
| Mental-health interviews                  | 21 interviews: 14 clients and 7 clinicians                                    | Local open-weight analysis of protected data                                                  | Interviewer/participant speech merged, creating an untested source confound ( <a href="#">Mathis et al., 2024</a> )                                                               |
| Japanese clinical interviews              | 30 patient/provider interviews at urban/rural hospitals                       | Direct cultural and low-frequency-theme test                                                  | One language/context/model; vendor data practices not reported ( <a href="#">Sakaguchi et al., 2025</a> )                                                                         |
| Firearm-violence survivor interviews      | 21 Black men; mean 11,503 words/interview; 118 experiments                    | Rare evidence on AAE, trauma, guardrails, and local models                                    | One region; open 1B/8B model family; human validation of all machine codes infeasible ( <a href="#">Zhu et al., 2026</a> )                                                        |
| Rohingya/host interviews                  | 2,407 interviews; 32,463 Q-A pairs; 19 binary codes                           | Participant-attribute error and downstream-estimate bias                                      | Primarily deductive binary coding after translation, not theme induction ( <a href="#">Ashwin et al., 2025/2026</a> )                                                             |
| Multi-condition health interviews         | 48 interviews across ILD, POTS, COPD                                          | Cross-dataset workflows with parent-ID provenance                                             | Consumer web models, one run/condition, one researcher handles output ( <a href="#">Bai and Finkelstein, 2026</a> )                                                               |
| Digital mental-health debriefs            | 99 transcripts; detailed comparison on 20 with 428 human excerpts             | Direct excerpt, code, and theme comparison of human, out-of-box, and knowledge-base workflows | Both LLM workflows miss 190/428 human excerpts; knowledge-base “quotes” include 68/251 summaries ( <a href="#">Parkington et al., 2025</a> )                                      |
| Patient consultation/reflexive interviews | 122 transcripts from 61 patients                                              | Human-led hybrid deductive-inductive comparison with quotation audit                          | Only moderate overlap; reported quotes are 68% verbatim, 20% paraphrased, 6% partial hallucination, 3% full hallucination, and 3% truncated ( <a href="#">Bang et al., 2026</a> ) |

| Dataset/domain cluster                    | Scale and analytic use                                                           | Evidence contribution                                                                                     | Main limitation                                                                                                                                                                |
|-------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Health-services implementation interviews | 167 interviews across 12 FQHCs; about 1.327 million words plus site-summary task | Researcher-primary task decomposition, quote-grounded retrieval, verification, and intervention synthesis | Preprint; RAG fragmentation causes tangential retrieval, lost perspective, and overgeneralization ( <a href="#">Ronaghi et al., 2026</a> )                                     |
| Longitudinal life narratives              | 154 analyzed interview sections; three open models × five seeds                  | Direct demographic-conditioning and expert fidelity study                                                 | Race/gender-conditioned semantic, affective, and thematic shifts; stereotyping and omission of deaths, violence, abuse, and addiction ( <a href="#">Subbiah et al., 2026</a> ) |
| Autistic/ADHD youth interviews            | 57 participants; 17,953 response documents reduced to 13,957; ten topics         | Same-corpus comparison of prior human RTA with topic modeling                                             | Masking, self-doubt, and strengths are underrepresented or absent; diagnosis clustering is confounded by interviewer identity ( <a href="#">Lukito et al., 2026</a> )          |
| Dutch nursing news                        | 317 articles; 31-item deductive validation subset; 275,878 words                 | Multilingual content-analysis comparison                                                                  | Models apply 181/183 codes versus 138 for humans, with low alpha and systematic overcoding ( <a href="#">Berkmortel et al., 2026</a> )                                         |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The AAOCA family is the most important dependency correction in the evidence base. LLM-TA, Auto-TA, SFT-TA, TAMA, and the 2026 refinement/provenance paper share the same nine focus groups and twelve published themes. Their pipeline comparisons are useful, but improvements cannot be counted as independent external validation. SFT-TA further converts ten of those themes into 300 paraphrased training examples before evaluating against the twelve-theme universe ([Yi et al., 2025](#)). The later provenance paper expands to five corpora, but its clinical human-theme comparisons still inherit AAOCA/SVCHD references and its best iteration is chosen using a composite on held-out test material ([Yi et al., 2026](#)).

Other dependence families also constrain apparent replication breadth. [NITA](#) extends GAITA on the same Down East corpus and question; [TA-MAS](#) and [Simon's human-versus-MAS comparison](#) share the Simon/CSCL survey and human analysis; the [QualiGPT preprint](#) and [later reliability study](#) form a dependent system family; and De Paoli's [Italian](#) and [saturation](#) studies extend a staged workflow. Several applied papers also reanalyze earlier human-coded corpora or inherit published themes. These records remain distinct when they contribute a new evaluation or workflow question, but they are not counted as independent dataset replications. The publication-family ledger in Appendix 19.2 preserves the preferred version and dependencies.

Language and demographic coverage remain uneven, not absent. The inspected corpus includes Finnish political text, Italian and German interviews, Dutch news, Japanese clinical interviews, translated Kenyan focus groups and Rohingya/host interviews, Ghanaian researcher deliberation, Korean/Chinese/Japanese social-media data, and English varieties including African American English. Kempny et al. nevertheless found that 93% of the 75 studies in their pre-May-2025 health-methods review used GPT-family models and identified limited non-English evidence ([Kempny et al., 2026](#)). The most diagnostic later studies suggest that ordinary English manifest content is the easier regime. Japanese culturally specific categories, translated refugee interviews, AAE, trauma narratives, facilitator speech, and rare clinical experiences produce qualitatively different errors ([Sakaguchi et al., 2025](#); [Ashwin et al., 2025/2026](#); [Zhu et al., 2026](#); [Hill et al., 2026](#)).

Specialized deductive tasks also resist transfer from general demonstrations. In a peer-reviewed ICPC 2026 study, four frontier/open models apply nine security-specific codes to 263 human experiment comments; even a refined prompt yields average chance-corrected performance in the limited range (GPT-5  $\kappa$ /adjusted accuracy .26 and Claude-4 .30), and added prompt detail does not produce a Bonferroni-significant improvement ([Camporese et al., 2026](#)). This is a coding task rather than theme induction, but it directly warns against claims that a generally strong model transfers to technical qualitative codes without domain validation.

## 8. Evaluation practices

### 8.1 Automatic metrics and their constructs

The central evaluation rule is to separate the construct, its operationalization, and evidence that the operation is valid. This follows the measurement framework of Jacobs and Wallach, which treats construct definition and measurement-model validity as prior to numerical comparison ([Jacobs and Wallach, 2021](#)).

| Operation                                 | Plausible construct                                                  | What it cannot establish alone                                               | Evidence needed before strong use                                                                                                         |
|-------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Exact/fuzzy match; ROUGE, BLEU, METEOR    | Surface similarity to a selected reference                           | Interpretive equivalence, evidence adequacy, reflexivity, uniqueness         | Human code/theme judgments, unmatched-output accounting, configuration sensitivity ( <a href="#">ROUGE</a> )                              |
| BERTScore/ cosine/learned semantic metric | Proximity in a model-specific representation                         | Same analytic claim, warrant, polarity, participant base, or theory          | Expert validation on code/theme pairs and split/merge sets; threshold transfer ( <a href="#">BERTScore</a> ; <a href="#">MoverScore</a> ) |
| One-to-one bipartite/Hungarian matching   | Parsimonious alignment at similar granularity                        | Legitimate many-to-one or one-to-many relations                              | Granularity-stratified analysis and unmatched false-positive penalties                                                                    |
| Many-to-many threshold matching           | Flexible recovery under splits/merges                                | Valid embedding space or threshold                                           | Threshold sensitivity, expert mapping, contradiction probes                                                                               |
| Reference coverage/recall                 | Recovery of content in one reference                                 | Content missed by all references; participant spread; interpretation quality | Multiple independent readings plus participant/evidence audit                                                                             |
| Novelty/divergence                        | Difference from a pooled/reference code space                        | Whether difference is insight or unsupported invention                       | Source fidelity and qualified human adjudication                                                                                          |
| Coherence/distinctiveness                 | Within-theme homogeneity/between-theme separation                    | Analytic depth; a tension-bearing theme may be heterogeneous                 | Expert ratings of central concept, boundaries, and counterevidence                                                                        |
| Repeated-run similarity                   | Procedural stability under tested variation                          | Validity; output can be stably shallow or biased                             | Crossed perturbations and substantive adjudication                                                                                        |
| Provenance/quote links                    | Traceability and auditability                                        | Correct, representative, sufficient interpretation                           | Exactness, entailment, omission, participant spread, and audit-usability tests                                                            |
| Cohen/Fleiss $\kappa$ , alpha, Gwet AC1   | Reliability of defined judgments under prevalence/design assumptions | Truth or quality across all traditions                                       | Units, prevalence, marginals, disagreements, expertise, intervals                                                                         |

| Operation    | Plausible construct                         | What it cannot establish alone | Evidence needed before strong use                                                 |
|--------------|---------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------|
| LLM-as-judge | Agreement with a judge's rubric/application | Independent human validity     | Cross-family judges, random order, repetition, expert calibration, error analysis |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

Parfenova et al.'s human study gives a compact example of metric–construct divergence. BERTScore favors closeness to the reference, yet three researchers evaluating fifteen items sometimes preferred human alternatives with greater analytic depth; Krippendorff's alpha was only .20 ([Parfenova et al., 2025](#)). HICode uses many-to-many matching at several cosine thresholds, a better topology for legitimate splits and merges than forced one-to-one matching, but two original annotators produced human precision/recall different from cosine results and agreed with alpha .31 ([Zhong et al., 2025](#)). Dörfel and Ammoneit's full-text comparison makes the divergence especially concrete: 13 of 14 LLM-derived codes cleared a .60 semantic-similarity threshold against human codes and repeated applications reached mean Jaccard .83, yet global coding-unit overlap with the human analysis was only .39 and the LLM code system lost domain-specific distinctions ([Dörfel and Ammoneit, 2026](#)). These are not reasons to abandon semantic matching; they are reasons to validate it on the intended interpretive task and at the unit where the analytic claim is made.

**Reviewer interpretation among the inspected full texts:** Chen et al. provide the most developed no-single-gold diagnostic in this evidence map. Four human and four differently prompted Gemma-3-27B coders generate codebooks for 127 community messages; an aggregate code space supports Coverage, Overlap, Novelty, and Jensen–Shannon Divergence, with ten runs used for stability. The authors explicitly frame the measures as a profile rather than a universal score. Their simulations also reveal failure: code flooding can improve Coverage, Overlap, and Novelty, while concepts omitted by every coder remain outside the aggregate space ([Chen et al., 2026](#)). The method replaces one gold codebook with a team-relative universe, an important but incomplete step toward plurality.

Among the inspected full texts, the July 2026 *Agreement Is Not Quality* preprint supplies a substantial source-symmetric verification. Five LLMs and three trained humans independently apply a 72-item, multi-label K–12 codebook to 2,560 messages. An independent domain expert, blind to source, judges 855 paired code sets. Mean human–LLM Jaccard is .30 versus approximately .52 human–human, yet among decisive human–LLM comparisons the expert prefers humans 51.5% and LLMs 48.5% ( $p=.537$ ); a Bradley–Terry model interleaves humans and models. This directly demonstrates that agreement and blind expert preference can diverge in both directions ([Liu et al., 2026](#)). Its boundary is equally important: one verifier, one codebook, one platform, a selected model set, and code assignment rather than inductive theme interpretation. **Reviewer interpretation:** a generic “agreement is not quality”

contribution is already occupied within this evidence map; the open advance is multi-expert, multi-domain, provenance-aware validation at code and theme levels.

The following full-text stress tests are especially informative because a favorable headline measure coexists with a failure on a different construct.

| Study                 | Favorable signal                                                                                               | Failure or boundary revealed                                                                                                                                                               | Supported inference                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Dörfel and Ammoneit   | 13/14 semantic mappings exceed .60; repeated-run Jaccard=.83                                                   | Global human-LLM coding-unit overlap=.39 and domain distinctions are lost                                                                                                                  | Semantic mapping and run stability do not establish applied interpretive equivalence ( <a href="#">official paper</a> ).                            |
| QualiGPT              | GPT self-agreement $\kappa$ =.87                                                                               | Human-human $\kappa$ =.73 but human-inclusive GPT agreement about .46; GPT-4o/Claude comparisons .38/.42                                                                                   | Same-model stability can exceed agreement with qualified humans because shared bias is stable ( <a href="#">official paper</a> ).                   |
| Parkington            | Two pipelines produce compact code/theme systems                                                               | Each misses 190/428 human excerpts; overlap is 238/428 and 124/428; 68/251 knowledge-base “quotes” are summaries                                                                           | Theme presence does not reveal excerpt omission or quotation alteration ( <a href="#">preprint</a> ).                                               |
| IDEAlign              | Best automated correlations with expert scores are .618/.683                                                   | Human-oracle correlations are .906/.812; style-only edits inflate embeddings by up to 82%                                                                                                  | Embedding proximity is partly a wording/style measure and requires expert construct validation ( <a href="#">preprint</a> ).                        |
| ICR                   | One configuration reaches cosine .89 and F1 .91                                                                | Human reliability is .86–1.00 versus LLM .35–.76; that high-score configuration has ICR .35                                                                                                | Lexical/semantic similarity can coexist with weak interpretive reliability ( <a href="#">preprint</a> ).                                            |
| Han LLM judge         | Best rank correlations are $\rho$ =.63 with accuracy and .67 with nuance/composite; averaged accuracy ICC=.782 | Individual-rater ICC=.492–.545; only 60/3,560 outputs are human-rated and model-level correlations use five points                                                                         | The judge recovers coarse model order better than individual interpretive magnitude ( <a href="#">preprint</a> ).                                   |
| Logan iterative/batch | Iterative theme-set Jaccard=.78/.86/.71 versus batch=.44/.71/.64                                               | Both omit lower-frequency fraud, safeguarding, risk-aversion, KYC, switching-barrier, or unique-selling-proposition concerns                                                               | Higher set overlap does not guarantee rare-theme survival ( <a href="#">official paper</a> ).                                                       |
| Misra local models    | Deductive prompting often improves context/theme-fit percentages                                               | Context/duplication/theme-fit remain 38.9/28.5/19.2 and 45.6/29.5/27.1 for Gemma; 22.9/16.3/11.7 and 43.7/36.2/26.8 for Llama; prose conflicts with the last table value by reporting 46.8 | Prompting regime changes results, but source context and thematic fit remain low and denominators require audit ( <a href="#">official paper</a> ). |

| Study                | Favorable signal                           | Failure or boundary revealed                                                                                                     | Supported inference                                                                                                                                  |
|----------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Li confidence study  | Models emit mean confidence .908/.922/.891 | Accuracy=.441/.520/.338; ECE=.467/.403/.553; a .95 anchor is adopted 96.5% and AUC collapses to .518; steering remains below .60 | Verbal confidence is not calibrated evidence and can become anchor following ( <a href="#">official chapter: 2026 conference/2027 proceedings</a> ). |
| Bang hybrid workflow | Every human theme appears at a broad level | Quotes are 68% verbatim, 20% paraphrased, 6% partially hallucinated, 3% fully hallucinated, and 3% truncated                     | Broad theme recovery can hide evidence corruption ( <a href="#">official paper</a> ).                                                                |
| Berkmortel           | Models recover plausible high-level themes | They apply 181/183 codes versus 138 for humans, with low alpha and systematic overcoding                                         | Coverage can be inflated by treating semantic presence as interpretive significance ( <a href="#">official paper</a> ).                              |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

Borchers et al. add a reporting caution: only 5/432 reported contrasts favor multi-agent consensus and 65 favor single agents, while the paper's stated 77,273 decisions does not reconcile with the 500,256 decisions implied by its described factorial. The comparative direction is informative; the unreconciled denominator limits exact aggregation ([official EDM paper](#)).

## 8.2 Human evaluation

Human protocols vary from one expert inspecting one transcript to blinded multi-analyst comparisons. TAMA's expert is clinically qualified but helped create the original reference themes; five lay reviewers are mentioned without enough procedure to support an independent qualitative-validity claim ([Xu et al., 2026](#)). SFT-TA uses four blinded raters—one surgeon, one qualitative analyst, and two laypeople—but does not report inter-rater reliability ([Yi et al., 2025](#)). Parfenova and Pfeffer correlate a composite metric with human scores on only ten items and do not report the number of experts ([Parfenova and Pfeffer, 2025](#)).

Later designs fill only parts of the gap. Reflexis observes twelve experienced researchers in six pairs, but two authors rate prompt relevance and there is no baseline ([Ye et al., 2026](#)). Han et al. calibrate an LLM judge on only 60 of 3,560 outputs; the five-model aggregate makes rank correlations unstable ([Han et al., 2026](#)). ICR uses trained human ratings to expose meaning-level disagreements hidden by cosine/F1, but a position paper and preprint family should not be treated as a settled universal metric ([Pérez et al., 2026](#); [ICR preprint](#)). Ronaghi et al. compare manual and LLM-assisted work before scale-up and retain contextual interpretation with researchers; its practitioner-usefulness evidence is stronger than pure

similarity, but the evaluators are the workflow's developers and the report remains a preprint ([Ronaghi et al., 2026](#)).

Hill et al. offer a stronger design: human analysts and ChatGPT-5, Claude 4 Sonnet, and QualiGPT were mutually blinded, outputs required quotations and segment identifiers, facilitator segments were excluded, and error was assessed under strict and broader definitions. Yet the study contains one two-hour focus group with seven adults, so the design is more persuasive than the generality ([Hill et al., 2026](#)). Zhu et al. partner with firearm-violence researchers, build the human codes through grounded-theory/inductive procedures, conduct member checking for the source analysis, and maintain local processing; however, the scale of machine outputs makes full human validation infeasible, motivating embedding thresholds and sampled audits ([Zhu et al., 2026](#)).

| Human-evaluation protocol      | Evaluators and independence                                                                            | Design strengths                                                                                         | Boundary for interpretation                                                                             |
|--------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <a href="#">TAMA</a>           | One clinically qualified expert helped create the reference analysis; five lay reviewers are mentioned | Expert stopping and one-transcript source audit                                                          | Reference dependence and underspecified lay procedure do not establish independent qualitative validity |
| <a href="#">SFT-TA</a>         | Four blinded raters: one surgeon, one qualitative analyst, two laypeople                               | Role-diverse blinded ratings                                                                             | No reported inter-rater reliability; raters evaluate inside the inherited AAOCA theme family            |
| <a href="#">Reflexis</a>       | Twelve experienced researchers in six pairs; two authors rate prompt relevance                         | Paired deliberation, logged disagreements, and positionality prompts                                     | No control baseline; author ratings and synthetic tasks bound outcome claims                            |
| <a href="#">Hill et al.</a>    | Human and model analyses mutually blinded                                                              | Source identifiers, quotation requirements, strict/broad error definitions, prevalence-aware reliability | One two-hour focus group with seven adults limits generalization                                        |
| <a href="#">Zhu et al.</a>     | Domain researchers develop the human analysis; member checking belongs to that source analysis         | Local processing, 118 experiments, source-grounded sampled audits                                        | Thousands of machine labels make exhaustive human validation infeasible                                 |
| <a href="#">Ronaghi et al.</a> | Workflow developers compare manual and assisted work; researchers retain final interpretation          | Task-specific usefulness, quotation verification, and observed intervention decisions                    | Developer evaluation and preprint status limit independence and maturity                                |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

A minimally adequate human evaluation should report evaluator count and expertise, independence from reference creation, blinding, randomized order, sampled units and rationale, rubric wording, independent versus consensus ratings, disagreement/reliability with uncertainty, statistical unit,

multiplicity, and time. The field's reporting problem is not unique: NLG reviews document inconsistent quality terms and frequent omissions of evaluators, sampling, rating design, and statistics ([van der Lee et al., 2019](#); [Howcroft et al., 2020](#)).

### 8.3 Cost and workload

Runtime is routinely compared with human coding hours, but verification is often excluded. QuaLLM reports 145.49 million tokens and a cost of \$1,662.30 ([Rao et al., 2025](#)). TAMA reports machine processing in minutes versus roughly 30 hours per human coder on AAOCA ([Xu et al., 2026](#)). Turner et al. estimate 97% time and 99% cost reductions for deductive dementia-trial coding, but 42.6% of 929 extracted excerpts were incorrect, making uncounted adjudication material ([Turner et al., 2026](#)). Ronaghi et al. report 30% and 55% time reductions for two researchers on the site-summary task and scale quote-grounded analysis to 167 interviews, while also documenting retrieval fragmentation, contextual correction, and continuing human synthesis ([Ronaghi et al., 2026](#)). Vikan et al.'s rigorous offline reflexive use found no time saving and documented fabricated quotations and changed meanings ([Vikan et al., 2025](#)). Efficiency claims should include cleaning, segmentation, prompt/codebook development, retries, source verification, false-positive review, privacy review, documentation, and reruns after model drift.

**RQ5 answer.** Evaluation ranges from exact/fuzzy matching, F1/accuracy,  $\kappa$ /alpha, lexical and embedding similarity, one-to-one or many-to-many matching, coverage/novelty/coherence, repeated-run stability, and LLM judging to blinded expert comparison, quotation audits, participant/source coverage, downstream estimates, workload, and cost. No single operation validates all qualitative claims. Adequacy depends on the declared construct, independent evaluators, plurality assumptions, source-level error categories, representative sampling, uncertainty and statistical units, calibration, and the verification work required to make outputs safe.

## 9. Validity, reliability, and interpretability

### 9.1 A reference interpretation is not objective ground truth

Human themes are evidence, but their epistemic role depends on the method. In coding-reliability designs, agreement with a carefully developed codebook can be the target. In reflexive TA, a reference is one situated reading. Several studies acknowledge this yet still organize quantitative evaluation around a consensus set. TAMA calls its twelve themes reliable ground truth while later noting that thematic interpretations are not unique ([Xu et al., 2026](#)). Dai's LLM participates in constructing the final codebook, so subsequent agreement is partly a compatibility test rather than independent discovery ([Dai et al., 2023](#)). A valid plural design should preserve alternative codes, rationales, and disagreements rather than collapse them before evaluation.

### 9.2 Reliability is conditional, not validity

Repeated-run similarity, inter-rater agreement, and consistent code assignment are valuable. They answer whether a procedure behaves similarly under specified conditions. They do not show that the stable output is warranted. Auto-TA's bidirectional ROUGE across ten runs can identify lexical instability, but cannot establish qualitative dependability ([Yi et al., 2025](#)). Bijker et al. obtain substantial agreement across multiple ChatGPT conversations in content analysis, yet that alone cannot reveal shared bias or systematic misreading ([Bijker et al., 2024](#)). Hill et al.'s 93% agreement paired with  $\kappa$  .34 illustrates prevalence sensitivity, while Gwet AC1 remained about .92-.93 ([Hill et al., 2026](#)). Reporting only the most favorable reliability coefficient invites misinterpretation.

Confidence is a separate construct and cannot repair this problem by assertion. In the AIED coding study, mean verbal confidence near .9 accompanies accuracy of .338-.520 and ECE of .403-.553; a .95 anchor is copied in 96.5% of outputs and leaves discrimination near chance (AUC .518) ([Li et al., 2026/2027](#)). Rationale similarity can prioritize disagreements for review, but Tajik et al.'s large rationale-similarity difference ( $d=1.16$ ) is a triage signal, not evidence that either agent is correct ([Tajik et al., 2026](#)).

### 9.3 Abstraction is not automatically quality

Systems often reward concise, nonredundant, semantically coherent themes. But high abstraction can erase cases, and embedding homogeneity can favor broad topics over interpretive themes that connect tension. In Zhu et al., BERTopic reduced thousands of labels to fewer than sixty, increasing apparent relevance but reducing formal-human-code capture and producing some tenuous, stereotypical, or unsupported labels ([Zhu et al., 2026](#)). In LLM-TA, a rare cardiac arrest can become disproportionately salient ([Raza et al., 2025](#)). Conversely, topic models of autistic/ADHD youth interviews underrepresented or missed masking, self-doubt, and strengths, and higher-Jaccard iterative analysis still missed low-

frequency concerns ([Lukito et al., 2026](#); [Logan et al., 2026](#)). Theme count, hierarchical depth, parsimony, and coherence should therefore be treated as design properties, not quality outcomes.

#### 9.4 Interpretability requires warrants

A theme name and several relevant quotations are inspectable, but an interpretive warrant must explain why those passages support that theme, how contrary passages were handled, and whose accounts are represented. A same-family judge can check surface consistency without revealing shared assumptions. TAMA's one-transcript GPT-4o-mini audit found no hallucinated themes and high grounding for GPT-4o output, but the authors acknowledge circularity and the limited audit ([Xu et al., 2026](#)). The 2026 trauma study is more revealing because source-linked examination differentiates likely valid novel codes from tenuous “reading between the lines” and unsupported stereotypes ([Zhu et al., 2026](#)).

## 10. Provenance, auditability, and reproducibility

Provenance has evolved from optional quotations to structured lineage. Official paper records and their linked artifacts document resources for Dai ([paper](#); [code](#)), LLoOM ([paper](#); [code](#)), HiCode ([paper](#); [code](#)), DeTAILS ([paper](#); [code](#)), CentaurTA ([paper](#); [code](#)), TAMA ([paper](#); [code](#)), LLM-TA ([preprint](#); [code](#)), Muse ([paper](#); [evaluation data](#)), and the trauma pipeline ([paper](#); [code](#)). **Reviewer judgment among the inspected full texts:** no single implementation dominates every provenance layer. The 2026 clinical refinement preprint carries stable turn IDs through chunks, quotations, normalized codes, typed relations, subthemes, themes, and LLM-reviewer revisions ([Yi et al., 2026](#)); Reflexis retains immutable code histories and disagreement/positionality events ([Ye et al., 2026](#)); Telescope provides branchable graph histories and analyst comparison ([Lee et al., 2026](#)); and ROCK specifies interoperable source spans, codebooks, relations, and versions ([Peters et al., 2026](#)).

Logging is not validation. The provenance paper uses GPT-4o-mini in generation and evaluation roles and reports no independent human validation of the ledger's interpretive warrants; public code, complete prompts, and generated ledgers were not located in the inspected version ([Yi et al., 2026](#)). Auto-TA has quote IDs but maps the fraction of used quotes to “credibility/confirmability,” which rewards reuse without checking representativeness ([Yi et al., 2025](#)). [Reflexis](#) validates prompt relevance and perceived control, not whether its histories improve evidential sufficiency; [Telescope](#) reports analyst confidence in saturation, not held-out coverage; [ROCK](#) is a specification rather than an adoption/outcome study. A theme can be perfectly traceable to one participant and still misrepresent the population.

A provenance-aware evaluation should test five layers separately: (1) **exactness**—does the cited string occur and retain speaker/context? (2) **entailment/warrant**—does it support the stated code/theme? (3) **coverage**—which relevant passages and participants are omitted? (4) **distribution**—is evidence concentrated in a few speakers/sites? and (5) **audit usability**—can an independent analyst reproduce the path, inspect versions, and register disagreement? Stratified human audits should sample high/low-frequency themes, negative cases, early/middle/late context positions, and every participant stratum.

Computational reproducibility remains limited by proprietary drift. Exact dated model identifiers, temperature, top-p, prompt text, context/chunking, retry policy, seeds or run IDs, tool versions, human edits, and stopping rules are often incomplete. Kempny et al. found that 45% of 75 studies did not specify deployment configuration, 75% reported no parameter settings, 61% supplied at least partial prompts, and 13% supplied no prompting details ([Kempny et al., 2026](#)). Restricted qualitative data can be ethically necessary, but papers can still release synthetic fixtures, schemas, prompt/configuration files, hashes, transformation logs, and a runnable audit harness.

## 11. Confounding, identity leakage, and shortcut learning

The expanded corpus contains direct confound and positionality evidence, so a claim that such tests are absent is no longer defensible. Lukito et al. find that apparent diagnosis clustering in autistic/ADHD interview topics is confounded by interviewer identity and is not robust after adjustment ([Lukito et al., 2026](#)). Subbiah et al. cross race/gender conditions, three open models, and five seeds on 154 life narratives and observe demographic-conditioned semantic, affective, and thematic shifts, stereotyping, and consequential omissions ([Subbiah et al., 2026](#)). Ashwin et al. regress coding errors on respondent attributes: GPT-3.5 errors are associated at the 5% level with characteristics for 10 of 19 codes, versus 1 of 19 for a bespoke classifier, and downstream regression estimates can be distorted ([Ashwin et al., 2025/2026](#)). Yu, Liu, and He find much higher within-company than cross-company similarity in earnings-call representations and strong alignment with a ticker-level historical baseline, showing how “text signal” can be repeated-source identity ([Yu et al., 2025](#)). These results demonstrate specific failure mechanisms; they do not prove shortcut use in every theme generator.

Several pathways are visible. Mathis et al. merge interviewer and interviewee speech, making facilitator language inseparable from respondent content ([Mathis et al., 2024](#)). Hill et al. explicitly exclude facilitator segments yet report that facilitator material drives much of the comprehensive error when systems mishandle source boundaries ([Hill et al., 2026](#)). Disease-background prompts in clinical studies can turn diagnosis language into a shortcut. Random chunk splits can leak the same participant or interviewer across train and test. Elangovan, He, and Verspoor show the general effect of input overlap: in SST-2, accuracy rises from 85.0 in the least-overlapping stratum to 96.8 in the most-overlapping stratum ([Elangovan et al., 2021](#)).

Adjacent task studies sharpen the mechanism. Rizvi et al. show that keyword and lexical cues are insufficient for nuanced ableism classification; Sakib et al. directly probe spurious correlations in social-determinants extraction; and Wei et al.'s China/Japan/Korea platform comparison entangles platform, language, geography, time, and preprocessing and may remove small minority clusters ([Rizvi et al., 2025](#); [Sakib et al., 2025](#); [Wei et al., 2026](#)). These are not interchangeable with inductive TA, but they justify subgroup, platform, source, and lexical-shortcut checks in a qualitative pipeline.

Long-context and order effects create a second confound. Evidence near the middle of long contexts can be used less reliably than evidence near the beginning or end ([Liu et al., 2024](#)). Few-shot exemplar order can change performance from near state of the art to near random, with good orders failing to transfer ([Lu et al., 2022](#)). A fixed temperature of zero does not control model drift, hidden serving nondeterminism, chunk order, exemplar order, or context position.

Contamination is an unresolved ambiguity when public corpora and published human themes are used with closed models. Deng et al.'s TS-Guessing method found that ChatGPT and GPT-4 exactly recovered

a missing MMLU option in 52% and 57% of cases, demonstrating that contamination can be probed rather than merely asserted ([Deng et al., 2024](#)). The AAOCA reference themes, source papers, and disease description are public. **Reviewer interpretation:** this creates a memorization risk, not evidence that the systems memorized them. A qualitative benchmark should document availability dates, use fresh/unpublished references where possible, mask disease/site/paper names, test paraphrased labels, insert secret placebos/canaries, and attempt missing-theme guessing before exposing the corpus.

The appropriate confound suite combines participant/interviewer/site/domain-disjoint folds; duplicate and nearest-neighbor analysis; probes for participant/source/site identity in embeddings; ablations of facilitator turns, boilerplate, demographics, disease names, and metadata; counterfactual identity swaps; chunk and codebook-order permutations; evidence-position randomization; and rare/contradictory-case insertion. Report theme quality alongside source-probe accuracy and participant-level residual error. A model that remains accurate only within source should be described as source-conditioned, not generally interpretive.

**RQ6 answer.** Demonstrated or credible threats include participant/interviewer/source identity, demographic conditioning, platform/language/time entanglement, overlapping participants or chunks, codebook/reference leakage, development-set reuse, prevalence and class imbalance, lexical style sensitivity, public-corpus contamination, long-context position and order effects, same-family judging, human anchoring, stable shared bias, majority dominance, and omission or sensational amplification of rare cases. Causal shortcut claims require controlled ablation or counterfactual evidence; otherwise the report labels them risks rather than facts.

## 12. Ethics and high-stakes clinical use

Ethical reporting is uneven even when papers discuss ethics in principle. Kempny et al. found that 95% of their broad sample mentioned ethical considerations and 97% incorporated some human verification, but these aggregate categories do not show whether consent covered vendor processing, whether retention was disabled, or who is accountable for an unsupported clinical theme ([Kempny et al., 2026](#)).

Local inference offers a concrete privacy path. Mathis et al. run a quantized LLaMA-2-70B model on local dual RTX 3090 hardware for protected mental-health interviews ([Mathis et al., 2024](#)). Vikan et al. use offline Mistral-7B, report ethics references and consent that explicitly covers LLM use, and retain human authorship/accountability ([Vikan et al., 2025](#)). Grover et al. use local Hermes-3-Llama-3.1-70B on five Canadian focus groups and keep data access under research-ethics control ([Grover et al., 2026](#)). Zhu et al. run local open models, withhold survivor transcripts, and manually check that released codes are not identifying ([Zhu et al., 2026](#)).

De-identification is not authorization and does not eliminate risk. Baroud et al. annotate 6,199 indirect identifiers across 100 MIMIC-III notes, showing how contextual combinations remain sensitive ([Baroud et al., 2025](#)). Morris et al. reconstruct 92% of 32-token inputs exactly from embeddings and recover 89% of full names from MIMIC-note embeddings, so hosted embedding APIs and vector stores require raw-text-level governance ([Morris et al., 2023](#)). A defensible paper should state the ethics determination, consent and secondary-use scope, whether LLM processing was disclosed, explicit and indirect de-identification, deployment and processor/subprocessor, region, encryption, retention, training use, deletion verification, access control, and incident response.

The expanded evidence also shows why provider assurances and reproducibility artifacts require inspection. Mehta et al. send translated, sensitive focus-group material through an API/Colab workflow; the paper's assurance about provider access is broader than what it documents ([Mehta et al., 2025](#)). Frontier privacy policies are fragmented and change over time, and King et al.'s prose generalization about default training is broader than its own table supports ([King et al., 2025](#)). The repository linked to Wei et al. prints an API key and contains language/label inconsistencies; public code therefore improves auditability only when secrets, configurations, and paper-code consistency are checked ([official paper](#); [repository](#)). Gustavsen et al. exclude a participant at reidentification risk while explicitly noting that this protection can silence a marginalized voice ([official paper](#)).

| Study/workflow                | Data governance and ethics                                              | Reproducibility provision                                      | Residual concern                                                                       |
|-------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <a href="#">Mathis et al.</a> | Yale exemption; local processing for protected mental-health interviews | Exact LLaMA-2-70B 4-bit/local stack and an accepted manuscript | Consent for model processing is not reported; merged speakers create a source confound |

| Study/workflow               | Data governance and ethics                                                                                            | Reproducibility provision                              | Residual concern                                                                                                            |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Vikan et al.</a> | Ethics/data-protection references, consent explicitly covering LLM use, encrypted consent, offline sensitive server   | Offline Mistral-7B workflow described                  | Data and reusable resources are necessarily restricted or not reported                                                      |
| <a href="#">Zhu et al.</a>   | IRB, payment, manual de-identification, university-local models, transcript withholding, released-code identity audit | <a href="#">Code</a> , prompts, and parameter tables   | Withheld data and sampled validation limit independent replication, appropriately trading access for participant protection |
| <a href="#">Mehta et al.</a> | Multiple approvals and written consent; translated sexual/reproductive-health material processed through API/Colab    | Annotated supplement, prompts/repeats, and dataset DOI | Provider-access assurance and the reported GPT-4o/version date are broader or less coherent than the documentation          |
| <a href="#">Wei et al.</a>   | Public posts; usernames removed; no IRB or consent reported                                                           | Public <a href="#">code/prompt repository</a>          | Repository prints an API key and conflicts with paper/file language-label semantics; full data are absent                   |
| <a href="#">King et al.</a>  | Public policy documents; substantive privacy analysis                                                                 | Sources and access dates reported                      | No archived policy snapshots or coded matrix; policies drift, and one prose generalization exceeds its table                |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

Representational harm is not hypothetical. In firearm-violence interviews, safety refusals are triggered by graphic injury, sexual activity, race, AAE, and a slur used non-derogatorily; the authors report an average refusal rate of 44% and up to 65% of some participants' experience ignored ([Zhu et al., 2026](#)). Japanese low-frequency and culturally specific content is unstable or absent ([Sakaguchi et al., 2025](#)). Ashwin et al. show that errors correlated with respondent characteristics can change policy-relevant estimates ([Ashwin et al., 2025/2026](#)). In high-stakes research, unsupported clinical specificity, minority erasure, and majority-pattern dominance should be treated as safety outcomes, not merely “model limitations.”

Human review is necessary but must be specified. A qualified qualitative analyst should own the interpretation; a domain expert should assess clinical meaning; an evaluator independent of the reference creators should audit evidence; and participant or stakeholder validation should be considered for purposively sampled claims. Disagreement and negative cases should remain visible. “Full automation” is not defensible for clinical interpretation on the reviewed evidence: local small models can refuse or stereotype, frontier APIs raise privacy and drift concerns, and even high agreement can coexist with source errors or low-prevalence failure.

Synthetic participants are not an uncomplicated privacy substitute. Kapania et al. show that LLM-generated “participants” simulate stories without the situated agency, accountability, and lived experience that qualitative research seeks ([Kapania et al., 2025](#)). They can support stress tests or synthetic fixtures, but should not silently replace people whose experiences ground the research claim.

**RQ7 answer.** The reviewed reports variously document ethics approval, consent, de-identification, local or enterprise deployment, restricted access, provider assurances, and code/data release, but reporting of consent for model processing, retention/training terms, embedding risk, subgroup harm, credential hygiene, stakeholder authority, and downstream accountability is inconsistent. In high-stakes work, privacy, refusal, omission, stereotyping, unsupported specificity, and exclusion-induced silencing should be evaluated as outcomes, with a named human owner able to halt or reject the analysis.

## 13. Critical synthesis

### 13.1 Points of agreement

Across the inspected architectures, clear definitions, inclusion/exclusion criteria, relevant examples, and bounded tasks generally improve code application. Muse's error analysis attributes much disagreement to under-specified codebooks and inconsistent human annotation; its best ensemble reaches  $\kappa$  .703 only on well-specified codes and after per-code tuning ([Matveyenko et al., 2026](#)). Dai's iterative rationales and examples produce high human-machine agreement on small survey tasks ([Dai et al., 2023](#)). HICode and LLoOM show that iterative abstraction scales better than one monolithic prompt ([Zhong et al., 2025](#); [Lam et al., 2024](#)). Evidence links and analyst-editable artifacts make outputs more inspectable and usable in these systems, even though they do not prove validity.

Ronaghi et al. sharpen the bounded-assistance conclusion in applied health services. Their model can organize researcher summaries and retrieve, verify, and sort quotation-grounded evidence across 167 interviews, making a deadline-constrained synthesis feasible; it cannot supply the actionable nuance, novelty judgment, contextual correction, or final intervention decisions, which remain human work ([Ronaghi et al., 2026](#)). NITA reaches a similar governance conclusion through a noncoding design, and Reflexis operationalizes optional disagreement and positionality support without replacing analyst authority ([NITA](#); [Reflexis](#)).

The inspected evidence also repeatedly finds that output quality is conditional on prompt wording, role, chunking, model, threshold, codebook definition, context, and run. Zhu et al. run 118 experiments and find large quality changes from small processing/prompt differences; model size changes relevance only modestly ([Zhu et al., 2026](#)). Muse finds eight of fifteen models outperform GPT-4o under its chosen prompt but recognizes that GPT-oriented optimization may favor nearby models ([Matveyenko et al., 2026](#)). TAMA's local Llama-3.1-8B evaluator approves poor themes early, while GPT-4o requires expert stopping because an automatic score plateaus and over-refinement appears ([Xu et al., 2026](#)).

### 13.2 Points of conflict

**Reviewer synthesis among the inspected full texts:** the sharpest conflict is between high semantic/reference alignment and qualitative adequacy. Short-label studies report respectable BERTScore, but human experts prefer deeper alternatives and disagree substantially ([Parfenova et al., 2025](#)). Dörfel and Ammoneit likewise obtain 13/14 clear semantic code mappings and mean run stability of Jaccard .83, while coding-unit overlap with the human analysis is only .39 and domain-specific distinctions are lost ([Dörfel and Ammoneit, 2026](#)). Clinical systems recover most reference themes, while audits find culturally specific omissions, dramatic rare-case overrepresentation, wrong source attribution, or narrative erasure ([Sakaguchi et al., 2025](#); [Raza et al., 2025](#); [Hill et al., 2026](#); [Zhu et al.,](#)

[2026](#)). These are not contradictory facts: a system can match the central, majority reference concepts while disagreeing at the applied unit or mishandling the edges where qualitative insight and ethical stakes are greatest.

A second conflict concerns automation. Thematic-LM and Auto-TA frame autonomous end-to-end generation as scalable; DeTAILS, Dai, LLoM, TAMA, Muse, and CentaurTA retain varying degrees of analyst control ([Qiao et al., 2025](#); [Yi et al., 2025](#); [DeTAILS](#); [Dai et al., 2023](#); [Lam et al., 2024](#); [Xu et al., 2026](#); [Matveyenko et al., 2026](#); [Wang et al., 2026](#)). The empirical evidence favors automation of bounded operations and human authority over interpretation. Role prompts labeled “qualitative researcher,” “surgeon,” or “layperson” do not create epistemic standpoint or accountability.

### 13.3 What apparent gaps are already partly closed

- **No-gold evaluation:** partly addressed by Chen's aggregate code space and Liu et al.'s blind, source-symmetric verification. Shared omissions, flooding, one-verifier dependence, theme-level warrants, and cross-domain construct validation remain ([Chen et al., 2026](#); [Liu et al., 2026](#)).
- **Provenance:** architecturally addressed by quote IDs, [clinical ledgers](#), [MindCoder](#), [ROSAnnotator](#), [Reflexis](#) histories, [Telescope](#) branches, and [ROCK](#)'s versioned standard. Exactness, sufficiency, omission, participant spread, and independent-auditor usability are not jointly validated.
- **Human feedback and reflexivity:** addressed by [Dai](#), [DeTAILS](#), [TAMA](#), [Muse](#), [CentaurTA](#), [Co-Refine](#), [NITA](#), [Nyaaba](#), [Reflexis](#), and the [K-12 instrument pipeline](#). Open questions concern anchoring, longitudinal outcome change, undetected error, burden, and whether deliberation improves independent interpretation rather than satisfaction.
- **Multi-agent analysis:** already crowded; novelty cannot rest on adding roles. Borchers' 5/432 versus 65/432 contrast directions make independent benefit especially doubtful. Agent identity, interaction, and advantage over strong sequential calls need ablation ([Borchers et al., 2026](#)).
- **Fine-tuning and adaptation:** demonstrated by [Parfenova et al.](#), [Ganesh et al.](#), [SFT-TA](#), and other LoRA/steering studies; held-out concept/domain/model-family generalization and leakage-safe adaptation remain open.
- **Grounded-theory and noncoding workflows:** schema/hierarchy systems, [LOGOS](#), [Neo-Grounded Theory](#), and [open-axial-selective instruments](#) exist; [NITA](#) demonstrates a deliberate noncoding alternative. Full theoretical sampling and validated theory development remain unestablished among inspected systems.
- **Stability and stopping:** repeated runs, no-new-theme rules, saturation ratios, fixed rounds, refinement trajectories, iterative-versus-batch comparisons, drift alerts, and analyst saturation

confidence all exist ([Aggregate Code Space](#); [De Paoli-Mathis](#); [QualiGPT](#); [Borchers et al.](#); [Logan](#); [Reflexis](#); [Teleoscope](#)). Crossed seed×prompt×order×sample×model×analyst uncertainty and a held-out stopping rule tied to supported minority/negative-case gain and verification burden remain open.

- **Confidence/abstention:** rationale and confidence routes exist, but severe miscalibration, anchor copying, and weak steering make an uncalibrated confidence threshold unsafe ([Li et al., 2026/2027](#)).
- **Multilingual and culturally situated work:** Finnish, Italian, German, Dutch, Japanese, translated East African, and East Asian platform studies exist ([Kousa et al.](#); [Sakaguchi et al.](#); [Mehta et al.](#); [Ashwin et al.](#)). Breadth, native-speaker evaluation, and cross-site/model transfer remain limited and uneven.

### 13.4 Overall evidentiary judgment

The one-to-one 12-dimension audit over all 181 core records yields the following distribution. Scores mean 0=absent/inadequate, 1=partial, and 2=adequate for the inspected report and claim; they are not additive study rankings.

| Quality dimension                   | 0  | 1   | 2   |
|-------------------------------------|----|-----|-----|
| Qualitative-method clarity          | 16 | 56  | 109 |
| Dataset appropriateness             | 16 | 60  | 105 |
| Computational-workflow transparency | 11 | 79  | 91  |
| Baseline strength/independence      | 46 | 78  | 57  |
| Human-evaluation adequacy           | 42 | 103 | 36  |
| Automatic-metric validity           | 62 | 84  | 35  |
| Interpretive plurality              | 78 | 69  | 34  |

| Quality dimension                    | 0  | 1  | 2  |
|--------------------------------------|----|----|----|
| Evidence grounding/traceability      | 50 | 66 | 65 |
| Stability/reproducibility testing    | 94 | 48 | 39 |
| Ethics/privacy/responsible reporting | 44 | 78 | 59 |
| Data/prompt/code/model availability  | 55 | 95 | 31 |
| Claim proportionality                | 22 | 78 | 81 |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The pattern shows comparatively frequent method descriptions and suitable datasets, but far fewer adequate scores for independent human evaluation, metric validity, plurality, stability, and reproducibility resources. Peer review does not remove these weaknesses, and preprint status does not automatically make a design uninformative. The critical unit is the claim: a code-application accuracy claim can be supported by a well-constructed reference and reliability study; a claim of reflexive or clinically meaningful interpretation requires stronger source, plurality, expert, and governance evidence.

**Reviewer interpretation:** current evidence justifies “LLMs can accelerate and scaffold specified operations in qualitative workflows, subject to human verification.” It does not justify “LLMs conduct equivalent reflexive thematic analysis” or “a high theme-overlap score establishes trustworthy clinical interpretation.” The most valuable next paper should stop optimizing plausible theme generation and instead experimentally validate who, what evidence, and which contradictions survive the pipeline.

## 14. Research-gap matrix

The matrix distinguishes an unresolved empirical claim from a feature that already exists. “Novelty risk” is reviewer risk after work inspected through 24 August 2026; “feasibility” assumes roughly seven weeks to the 12 October ARR deadline. Every absence statement is **reviewer judgment among the inspected full texts**, qualified by 53 unique reports that could not be retrieved and by focused rather than exhaustive citation chaining.

| Gap                                                   | Papers that partly address it                                                                                                                                                                                                                                                                                                                                                                                                   | Why it remains unresolved                                                                                                                                                                                             | Evidence needed                                                                                                                                            | Novelty risk                                                                 | Feasibility by 12 Oct                            | Likely ARR significance | Recommend ed design                                                                                                                                                                                      |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participant, minority, and negative-case preservation | LLM-TA shows rare-event amplification ; Sakaguchi and Zhu show cultural/trauma omission and refusal; Parkington misses 190/428 excerpts; Subbiah shows demographic-conditioned distortions; Lukito and Logan show implicit/rare-theme loss ( <a href="#">LLM-TA</a> ; <a href="#">Sakaguchi</a> ; <a href="#">Zhu</a> ; <a href="#">Parkington</a> ; <a href="#">Subbiah</a> ; <a href="#">Lukito</a> ; <a href="#">Logan</a> ) | No inspected study jointly measures speaker-level evidence precision/recall, concentration, contradiction survival, rare-case calibration, source shortcuts, and unsupported synthesis under controlled perturbation. | Stable source/participant IDs; controlled prevalence/position/remove/swap tests; independent warrant and omission judgments; downstream conclusion change. | Low–medium only for the <b>joint</b> audit; high for a simple coverage score | High with existing authorized multi-speaker data | High                    | Exact source ledger; participant/interviewer/site-disjoint conditions; rare/negative-case sentinel set; blinded qualitative and domain experts; report label-, evidence-, and conclusion-level failures. |

| Gap                                                  | Papers that partly address it                                                                                                                                                                                                                                                                                         | Why it remains unresolved                                                                                                                                                                                                                   | Evidence needed                                                                                                                                                                               | Novelty risk                                                                             | Feasibility by 12 Oct         | Likely ARR significance                     | Recommended design                                                                                                                                                                      |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity/<br>source/<br>domain<br>confounding        | Yu demonstrate source identity; Ashwin demographic residual error; Lukito interviewer confounding; Subbiah demographic conditioning; Rizvi/Sakib lexical shortcuts ( <a href="#">Yu</a> ; <a href="#">Ashwin</a> ; <a href="#">Lukito</a> ; <a href="#">Subbiah</a> ; <a href="#">Rizvi</a> ; <a href="#">Sakib</a> ) | Direct evidence now exists, but causal ablations, theme-level consequences, and transfer across sources/models remain sparse.                                                                                                               | Disjoint folds, identity probes, cue ablations and counterfactual swaps, interaction effects, residual error by participant, held-out sites/models.                                           | Medium; “confounds may exist” is occupied                                                | Medium–high if metadata exist | High                                        | Cross participant×interviewer×site×domain design; mask facilitator/boilerplate/disease/site cues; measure probe accuracy and source-grounded theme quality together.                    |
| Plural,<br>source-symmetric<br>warrant<br>validation | Aggregate Code Space, <i>Agreement Is Not Quality</i> , IDEAlign, ICR, Han, Simon, Misra, and Logan address pieces ( <a href="#">Chen</a> ; <a href="#">Liu</a> ; <a href="#">IDEAlign</a> ; <a href="#">ICR</a> ; <a href="#">Han</a> ; <a href="#">Simon</a> ; <a href="#">Misra</a> ; <a href="#">Logan</a> )      | ACS is team-relative/flooring-sensitive; blind verification uses one expert/code book/domain; ICR and judge calibration remain narrow; theme-level alternative readings and source sufficiency are not validated across analytic contracts. | Multiple independent interpretations and experts; code- and theme-level items; source warrants/counter-evidence; held-out domain and model family; calibration and generalizability analysis. | Medium–high for generic “agreement is not quality”; medium for cross-contract validation | Medium                        | High if genuinely multi-expert and held out | Compare lexical, embedding, one/many-to-many, ACS, ICR, LLM judges, source-symmetric preference, and participant coverage against blinded expert warrants while retaining disagreement. |

| Gap                                                   | Papers that partly address it                                                                                                                                                                                                                                  | Why it remains unresolved                                                                                                                                                         | Evidence needed                                                                                                                                                           | Novelty risk                                                         | Feasibility by 12 Oct                   | Likely ARR significance                        | Recommended design                                                                                                                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provenance-aware validity, not another ledger         | <a href="#">Clinical ledgers</a> , <a href="#">MindCode</a> , <a href="#">ROSA</a> , <a href="#">NITA</a> , <a href="#">Nyaaba</a> , <a href="#">Reflexis</a> , <a href="#">Teleoscope</a> , and <a href="#">ROCK</a> implement links, histories, or standards | Exactness, context preservation, entailment, sufficiency, omission, participant spread, counterevidence, and independent-auditor usability are not jointly validated.             | Gold source offsets plus expert warrant/omission labels; planted attribution/context errors; auditor completion, time, disagreement, and version-reconstruction outcomes. | High for interface/schema alone; medium for validated audit outcomes | High as an evaluation layer             | High                                           | Implement or reuse ROCK-like lineage; compare no-ledger, quote-only, and versioned graph; stratify audits by rarity, participant, context position, and disagreement.             |
| Calibrated confidence, abstention, and review routing | Zhao-Liu confidence/diversity and Tajik rationale triage propose signals; Li documents overconfidence, anchoring, and failed steering ( <a href="#">Tajik</a> ; <a href="#">Li</a> )                                                                           | Verbal confidence is badly miscalibrated; rationale disagreement is not correctness; benefits under model/domain in shift and effects on minority cases/human burden are unknown. | Held-out calibration, per-code reliability, selective-risk/coverage curves, rare-case audits, model/domain in transfer, correction-time study.                            | Medium; generic confidence paper is crowded                          | High computationally; medium with users | Medium-high as part of WarrantBench/VoiceTrace | Compare logits where available, verbal confidence, ensembles, rationale disagreement, conformal/selective baselines, and human review; freeze thresholds before held-out domains. |

| Gap                                                   | Papers that partly address it                                                                                                                                                                                                                              | Why it remains unresolved                                                                                                                                    | Evidence needed                                                                                                                                                    | Novelty risk                     | Feasibility by 12 Oct | Likely ARR significance                  | Recommended design                                                                                                                                              |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cross-domain/<br>model/<br>temporal<br>generalization | <a href="#">HiCode</a> and <a href="#">Muse</a> span datasets; <a href="#">Than</a> and <a href="#">Ganesh</a> compare models/adaptation; <a href="#">Campores</a> shows technical-domain failure; <a href="#">Subbiah</a> crosses three models/five seeds | Prompts and thresholds are often tuned in-domain; proprietary snapshots drift; human validation is rarely crossed across sites, languages, models, and time. | Frozen prompts/configurations, family-disjoint judges, site/participant splits, temporal snapshots, multilingual/native-speaker validation, complete denominators. | Medium                           | Medium                | Medium-high                              | Predeclare model/prompt before held-out domains; cross open/proprietary families and versions; report interactions and error taxonomy, not pooled average only. |
| Stability as substantive uncertainty                  | <a href="#">Chen</a> , <a href="#">Auto-TA</a> , <a href="#">Jain</a> , <a href="#">De Paoli</a> , <a href="#">Mathis</a> , <a href="#">QualiGPT</a> , <a href="#">Borchers</a> , and <a href="#">Logan</a> repeat runs or compare processing modes        | Most vary one factor and score lexical/set similarity; stable shared bias and rare-case loss remain possible.                                                | Crossed seed×prompt×order×chunk×sample×model×analyst perturbations; variance decomposition at evidence/the conclusion levels; no best-run selection.               | High if merely rerunning prompts | High                  | Medium, high when tied to voice/warrants | Fractional factorial design with intervals; adjudicate cases whose evidence or substantive conclusion changes, not just labels.                                 |

| Gap                              | Papers that partly address it                                                                                                                                                                                                                                                                                                                                                                                                     | Why it remains unresolved                                                                                                                                                                                    | Evidence needed                                                                                                                              | Novelty risk                                        | Feasibility by 12 Oct                     | Likely ARR significance                 | Recommended design                                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Held-out evidence-based stopping | <a href="#">CAILA</a> no-new-theme, <a href="#">De Paoli-Mathis</a> saturation, <a href="#">TALLMesh</a> allocation, <a href="#">LOGOS</a> trajectories, <a href="#">LLMCode</a> iterations, <a href="#">TAMA</a> and <a href="#">clinical-refinement</a> scores, <a href="#">CentaurT</a> A degradation, <a href="#">Teleoscope</a> saturation confidence, and <a href="#">Logan</a> iterative/batch comparisons provide signals | No inspected rule jointly balances marginal supported evidence, participant/minority recall, negative cases, redundancy, unsupported claims, instability, and verification burden and then transfers unseen. | Complete refinement trajectories; held-out domains/models; independent adjacent-iteration judgments; predeclared rule and Pareto components. | Medium; simple saturation/no-new-theme is high-risk | High computation ally; medium for experts | Medium-high if transfer is demonstrated | Compare fixed rounds, no-new-theme, saturation ratio, human stop, same/cross-family judge, similarity plateau, batch/iterative, confidence, and frozen evidence-gain rule. |

| Gap                                                                   | Papers that partly address it                                                                                                                                                                                                                                              | Why it remains unresolved                                                                                                                                                       | Evidence needed                                                                                                                                                                                | Novelty risk | Feasibility by 12 Oct                         | Likely ARR significance | Recommended design                                                                                                                                                              |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prospective workflow value and accountability                         | <a href="#">DeTAILS</a> , <a href="#">Muse</a> , <a href="#">CentaurTA</a> , <a href="#">LLMCode</a> , <a href="#">NITA</a> , <a href="#">Nyaaba</a> , <a href="#">Reflexis</a> , <a href="#">Ronaghi</a> , and <a href="#">K-12 decision logs</a> show real interventions | Short formative studies, author evaluation, or preference do not establish reduced error, better depth, less anchoring, longitudinal learning, or changed decisions.            | No-AI/ constrained-assistant/ autonomous-draft arms; pre/post interpretations; errors, overrides, time by stage, confidence, learning, and consequential decisions.                            | Medium       | Low-medium unless recruitment/ data are ready | High                    | Prospective preregistered field or high-fidelity lab study with versioned logs, independent audits, and authority to reject/backtrack/stop.                                     |
| Independent quotation integrity and clinically meaningful specificity | <a href="#">TAMA</a> , <a href="#">Hill</a> , <a href="#">Turner</a> , <a href="#">Zhu</a> , <a href="#">Parkington</a> , <a href="#">Bang</a> , <a href="#">Mehta</a> , <a href="#">Cook</a> , and <a href="#">Gustavsen</a> audit different source errors                | Definitions and denominators differ; same-family judging persists; exact quotations can be miscoded; omission and clinically actionable specificity are rarely judged together. | Predefined fabricated/altered/paraphrased/composite/truncated/misattributed/unsupported/overgeneralized/omitted categories; blinded domain+method experts; severity and denominator reporting. | Medium       | Medium                                        | High in clinical NLP    | Exact-string and source-offset checks plus contextual warrant review; adversarial placebos; sample high/low-frequency and high-consequence claims; include verification burden. |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The novelty boundary is now narrow. Multi-agent roles, semantic overlap, quote/provenance ledgers, reflexivity prompts, repeated runs, confidence scores, and simple saturation/stopping rules already exist. A competitive contribution must validate a construct, intervention, or failure mechanism against those baselines and show why the result changes a qualitative conclusion or human decision.

## 15. Recommended ARR research direction

The ranking is **reviewer judgment**, not a result of the quality-score totals. It weighs novelty after the inspected literature, ACL relevance, ethical importance, and seven-week feasibility. All three projects require a frozen protocol, exact model/configuration records, repeated runs, an explicit analytic contract, and source-grounded human evaluation.

### Rank 1 — VoiceTrace: participant- and source-aware validity for LLM thematic analysis

**Provisional title.** *Whose Voice Survives? Participant, Source, and Negative-Case Audits for LLM-Assisted Thematic Analysis*

**Central claim.** Conventional theme similarity can remain high while a pipeline relies on source/identity cues or loses, misattributes, stereotypes, and overamplifies participant evidence. A joint provenance- and-perturbation audit reveals these failures and changes system rankings or substantive conclusions.

**Research questions.** (1) How much code/theme variation is explained by participant, interviewer, site, identity condition, and context position? (2) Which participant-coverage, concentration, counterevidence, and source-warrant measures expose failures hidden by semantic overlap and LLM judging? (3) Which chunking, masking, retrieval, model-family, abstention, and human-review choices preserve rare and contradictory accounts?

**Novelty relative to inspected work.** Provenance, demographic sensitivity, and reflexivity are each occupied. [Subbiah](#) directly demonstrates demographic-conditioned narrative shifts; [Lukito](#) shows interviewer confounding; [Parkington](#), [Bang](#), [Mehta](#), and [Gustavsen](#) expose omitted or altered evidence; [MindCoder](#), [ROSAnnotator](#), [clinical ledgers](#), [Nyaaba](#), [Reflexis](#), [Teleoscope](#), and [ROCK](#) provide lineage or deliberation components; [NITA](#) preserves researcher authority. The defensible novelty is the **joint controlled validation** of exact source lineage, participant/source perturbations, voice and negative-case preservation, and independent warrants—not a new ledger or dashboard.

**Datasets required.** Two multi-speaker corpora with stable participant/source IDs and authorization for model processing, ideally one approved clinical or health-services corpus and one public nonclinical corpus; one site/domain fully held out. If fresh clinical authorization is not already complete, use an approved existing corpus plus a public set and a human-checked synthetic perturbation suite. Do not use AAOCA as the sole test family.

**Experimental design.** Freeze prompts before held-out evaluation. Cross at least one strong proprietary and one open-weight model with chunking, source masking, evidence retrieval, and three independent runs. Use participant/interviewer/site/domain-disjoint partitions; source-identity probes; removal of facilitator, boilerplate, diagnosis, and site cues; counterfactual identity swaps; rare/negative-case

insert/remove/prevalence tests; context-position moves; and no-source-ID versus exact-source-ledger ablations. Report exact evidence precision, attribution error, participant recall/coverage/concentration, negative-case survival, unsupported-claim rate, conventional one/many-to-many similarity with threshold sensitivity, and at least one downstream conclusion change.

**Human evaluation required.** Separate reference creators from evaluators. Use at least two qualitative-method experts and two domain experts, blinded to condition and source, with independent judgments before adjudication. Stratify by participant prevalence, disagreement, negative-case status, source position, and model confidence. Ask whether the cited evidence supports the claim, what relevant evidence is omitted, and whether counterevidence changes the interpretation. Add purposive participant/stakeholder review for high-stakes claims if permitted.

**Ethics requirements.** Confirm consent and secondary-use scope for model processing; use local or approved enterprise deployment; document retention/training terms and embedding risk; suppress identifying quotations; preregister subgroup analyses; involve community/domain advisers in sentinel construction; and name the human who owns and can stop the interpretation.

**Key baselines.** Human-only analysis; single-pass/no-lineage generation; a codebook assignment baseline; HICode-style hierarchy; Thematic-LM/Auto-TA-style role pipeline; ROCK-compatible exact-source ledger; Reflexis-style deliberation; NITA-style researcher-primary condition; Subbiah-style demographic conditioning; no-source-ID ablation; and calibrated versus uncalibrated review routing.

**Expected contribution.** A benchmark and protocol, participant/source-aware metrics, controlled causal probes, and evidence that common rankings or conclusions change when voice preservation is measured. **Major risk:** unavailable multi-site data or insufficient expert time. **Mitigation:** freeze a two-corpus minimum, preregister the smallest defensible expert sample, release the perturbation/audit harness, and scope clinical claims to authorized data. **October feasibility:** high only if data and experts are secured immediately; otherwise medium with a nonclinical main experiment.

## Rank 2 — WarrantBench: plural, source-symmetric evaluation of codes and themes

**Provisional title.** *Beyond Agreement: Evaluating Competing Qualitative Interpretations Through Evidence Warrants*

**Central claim.** A multi-expert, source-grounded protocol that preserves alternative readings, counterevidence, and participant coverage can identify which automatic measures are useful under different analytic contracts without assuming one correct interpretation.

**Research questions.** (1) How do lexical, embedding, matching, plural-code-space, ICR, LLM-judge, blind-preference, provenance, and coverage measures relate to expert judgments at code and theme levels?

(2) Do relationships change under coding-reliability, codebook, and reflexive contracts? (3) Which metric portfolio transfers to an unseen domain and model family while detecting unsupported or unrepresentative interpretations?

**Novelty relative to inspected work.** [Aggregate Code Space](#), [source-symmetric Bradley-Terry comparison](#), [IDEAlign](#) perturbations, [Simon's blinded comparison](#), [specialist/lay clinical ratings](#), [ICR](#), [Han's LLM-judge calibration](#), [Misra's context/theme-fit audit](#), and [Logan's iterative/batch comparison](#) already cover parts of the problem. Novelty must be **theme-level, multi-expert, cross-contract, source-grounded construct validation on held-out domains/models**, not another scalar metric or “agreement is not quality” slogan.

**Datasets required.** Three domains with source IDs and at least two genuinely independent human interpretations per domain; a minimum viable version can use two deep domains plus a held-out transfer set. Include different qualitative contracts and purposively sampled minority/negative cases.

**Experimental design.** Generate outputs from two model families and two workflows. Build balanced code/theme comparisons covering paraphrase, split/merge, contradiction, generic abstraction, unsupported novelty, shared omission, source misattribution, and minority evidence. Experts blind to source judge central claim, evidence sufficiency, counterevidence, participant spread, alternative defensibility, and contract fitness. Fit hierarchical/generalizability models with evaluator/domain effects; hold out one domain and one model family; publish calibration, intervals, and error strata rather than one leaderboard.

**Human evaluation required.** At least three qualified experts per domain or a crossed incomplete-block design supporting reliability and generalizability estimates. Collect rationales and positionality information, retain individual disagreement before adjudication, randomize order, and disclose sampling and multiplicity.

**Ethics requirements.** Fair compensation; consent-compatible source quotation use; de-identification and access control; no unnecessary source-identity exposure; respectful disagreement-preserving release; and stakeholder input on harms and rubric language in sensitive domains.

**Key baselines.** ROUGE/METEOR; BERTScore/cosine; one-to-one and many-to-many matching; Aggregate Code Space dimensions; IDEAlign; ICR; repeated-run similarity; same- and cross-family LLM judges; source-symmetric Bradley-Terry; human-oracle/reference creators; quote exactness and participant coverage. No LLM judge may be the sole verifier.

**Expected contribution.** An empirical construct map showing which metrics predict independently warranted interpretation under which contract, plus a reusable judgment set. **Major risk:** annotation burden and the objection that judgments remain situated. **Mitigation:** preregister constructs, use a

powered incomplete-block design, preserve rationales/disagreement, and test generalization rather than claiming a universal gold metric. **October feasibility:** medium; two rigorous domains are preferable to three shallow ones.

### Rank 3 — StopTA: held-out evidence-based stopping for iterative refinement

**Provisional title.** *When Should an Agent Stop Refining? Evidence-Gain Criteria for Iterative Thematic Analysis*

**Central claim.** Fixed rounds, no-new-theme rules, similarity plateaus, confidence, and judge scores do not reliably identify improvement. A frozen rule balancing marginal supported participant/negative-case evidence against redundancy, unsupported claims, instability, and verification burden transfers better across models and domains.

**Research questions.** (1) What changes over refinement trajectories at evidence, participant, theme, and conclusion levels? (2) Which stopping rule transfers across models, domains, and pipelines? (3) Does exposing a stopping signal help analysts avoid over-refinement without anchoring or hiding minority cases?

**Novelty relative to inspected work.** [CAILA](#) implements no-new-theme stopping; [De Paoli-Mathis](#) estimate saturation; [TALLMesh](#) allocates transcript batches; [LOGOS](#) and [LLMCode](#) retain refinement trajectories; [TAMA](#) shows plateau and expert stopping; [clinical refinement](#) runs up to ten rounds; [CentaurTA](#) can degrade; [Teleoscope](#) captures analyst saturation confidence; [Logan](#) directly compares iterative and batch modes. The open claim is transferable held-out stopping with a joint evidential objective, not invention of a stopping signal.

**Datasets required.** Three existing public or approved corpora with stable source IDs, including at least one with rare/negative cases and one held-out domain. Save every intermediate artifact and human revision.

**Experimental design.** Produce complete ten-round trajectories for at least two models and two pipelines. Learn or calibrate only on development corpora, freeze the rule, and test on the third. Evaluate marginal supported evidence, participant/negative-case coverage, redundancy, unsupported claims, substantive instability, and verification time. Report Pareto fronts and component ablations; never select the best round on test material.

**Human evaluation required.** Blinded experts compare adjacent iterations and stopped outputs for supported improvement, lost nuance, counterevidence, minority preservation, and verification burden. A small analyst study can test whether a signal changes stopping behavior and whether it anchors users.

**Ethics requirements.** Apply the same source/privacy controls as Rank 1; recognize that intermediate artifacts can retain sensitive quotations; minimize retention; and make the analyst—not the stopping score—the final authority.

**Key baselines.** Fixed 1/3/5/10 rounds; no-new-theme; saturation ratio; TALLMesh allocation; iterative versus batch; same- and cross-family judge thresholds; reference-similarity plateau; human stop; Telescope-style saturation confidence; confidence/abstention; and Pareto-front reporting.

**Expected contribution.** Public stopping trajectories, a frozen rule, and evidence that “more refinement” is not monotonic and that the chosen stopping point transfers. **Major risk:** a hand-weighted composite that overfits. **Mitigation:** preregister components, fit no weights on test data, compare Pareto alternatives, and validate with independent human trajectories. **October feasibility:** high computationally and medium for the required expert study.

## Recommendation

Choose Rank 1 if participant/source identifiers, processing authorization, and evaluator access are already secured. **Reviewer judgment among the inspected full texts:** it offers the strongest combination of novelty, ACL relevance, and ethical consequence because it joins mechanisms that existing studies examine separately. Choose Rank 3 if those prerequisites are unavailable within days; it is more feasible but must demonstrate held-out transfer. Rank 2 is potentially the most foundational, but the existing no-gold and judge-calibration literature makes a rushed single-domain or single-expert version unlikely to clear the novelty bar.

## 16. Limitations of the review

This is a single-reviewer, Codex-assisted scoping review, not an independently duplicated screen or extraction. All 1,360 deduplicated database records now have terminal title/abstract dispositions, but terminal does not mean error-free: the initial 122/612/626 high/uncertain/low triage was only a prioritization device, and the later eligibility decisions were not independently adjudicated. Parallel extraction, schema assertions, DOI/title deduplication, evidence locations, and record-specific reasons improve auditability but do not create inter-reviewer agreement.

Retrieval was incomplete. The primary unique-identity flow sought 317 reports, did not retrieve 53, and assessed or resolved 264; the latter produced 237 retained reports, 18 full-text exclusions, and nine publication/version-family resolutions. No methodological or result claim in this synthesis derives from a nonretrieved report or publisher abstract. The appendix gives every nonretrieval its recorded retrieval disposition and every full-text exclusion its record-specific reason; four nonretrieval source records lack a specific reason and are labeled only as not retrieved after documented attempts rather than being supplemented by inference. Because plausible full texts remain inaccessible, all novelty and absence claims are bounded to reviewer judgment among the inspected full texts.

Coverage also limits recall. PubMed and arXiv supplied reproducible exports, but Crossref searches were relevance-ranked and capped, the arXiv identity query was capped, ACL retrieval was title-oriented, and OpenReview candidates were client-filtered. Stable auditable bulk exports were unavailable for Google Scholar, Semantic Scholar, ACM DL, IEEE Xplore, Web of Science, and Scopus in the available environment. Citation chaining preserved parent-seed provenance but was focused and index-dependent; no exhaustive saturation claim is made.

Publication status near the cutoff is unusually fluid. Preprint, online-first, acceptance, event, and formal-proceedings years can differ. Auto-TA is arXiv:2506.23998v2 with OpenReview label [ACL-SRW 2025 Poster](#); no archival ACL Anthology paper was verified ([preprint](#); [OpenReview history](#)). SFT-TA is arXiv:2509.17167v1 with OpenReview label [ML4H 2025 RejectfromProceedingsasFindingsPoster](#); its [PMLR XXX](#) text is not an archival citation ([preprint](#); [OpenReview history](#)). The retained AIED confidence paper was available as a 2026 conference manuscript but has a formal Springer 2027 proceedings citation and is labeled accordingly ([Li et al.](#)). The ATA-LLM paper was associated with the SEET 2025 event and published in 2026 ([Beltrán et al.](#)). Where peer review or acceptance was not established by 24 August 2026, the bibliography says so rather than inferring status from a DOI or template.

The corpus is heterogeneous and dependence-rich, so no meta-analysis was attempted. Identically named metrics use different formulas; coding units and denominators differ; studies reuse datasets,

human themes, prompts, or preprint/VOR families; and private data prevent independent source verification. Quality scores are reviewer judgments reported by dimension, not a total-score ranking. The 34-record method-congruence audit uses an explicit strict lower-bound rubric, but other qualified reviewers could reasonably code ambiguous traditions differently.

Model evidence is time-dependent. Proprietary snapshots, hidden serving changes, prompts, chunking, retries, seeds, and evaluator configurations are frequently incomplete. Citation counts are dynamic and retained only with their source and retrieval date in the master. Finally, the ARR recommendation depends on local facts the literature cannot provide—data authorization, compute, evaluator availability, stakeholder access, and engineering readiness—so its feasibility ranking must be revised if those prerequisites differ.

## 17. Conclusion

LLMs now assist nearly every visible artifact of qualitative work: familiarization proxies, quotation retrieval, open and deductive codes, codebooks, hierarchies, noncoding themes, memos and schemas, reports, disagreement/confidence triage, stopping signals, evidence links, version histories, and standards. **Reviewer synthesis among the inspected full texts:** the most defensible systems keep these artifacts editable and traceable, preserve analyst authority and disagreement, declare the analytic contract, calibrate uncertainty, and audit sources and participants with independent human judgment.

The evidence is strongest for well-specified code application, retrieval, and scalable organization. It is weaker—not wholly absent—for reflexive, culturally situated, minority-sensitive, theory-building, and clinically consequential interpretation. New direct evidence makes the boundary clearer: high semantic similarity, run agreement, topic coherence, or verbal confidence can coexist with low coding-unit overlap, quote alteration, systematic overcoding, rare/implicit-theme loss, demographic distortion, and interviewer confounding. A source link improves auditability but does not prove that an interpretation is exact, sufficient, representative, or warranted.

**RQ8 answer.** The open research opportunity is no longer “generate themes with agents,” “add provenance,” “prompt reflexivity,” “repeat runs,” or “stop when no new themes appear.” Those components exist. The strongest defensible ARR direction is a joint participant/source-aware validity audit that controls identity and context while measuring exact evidence, omission, concentration, negative-case survival, independent warrants, and changed conclusions. Multi-domain multi-expert warrant validation and held-out evidence-based stopping remain credible alternatives; calibrated abstention is an important component, not a substitute for either. These are reviewer judgments bounded to 24 August 2026, the inspected full texts, and immediate feasibility constraints.

## 18. Complete references

This bibliography contains every record in the frozen evidence master: **237 unique publication identities = 181 core reports + 56 adjacent, contextual, or methodological records**. Each evidence ID and frozen official URL appears once. Bibliographic fields start from the verified master; transparent enrichments correct the malformed Braun–Clarke and Elangovan titles, the Braun–Clarke venue, the official Jowsey and Montes titles, and missing QualAnalyzer authors, expand verified author lists where local union or citation metadata support them, and state the exact OpenReview-derived nonarchival type/status histories for Auto-TA and SFT-TA. **Et al.** remains for 14 records whose frozen local sources do not contain the complete list. The additional publications and verification artifacts below are cited in the narrative but are not added to the 237-record evidence count.

### 18.1 Core reports (181)

- **ta\_0077\_2026\_wahbeh.** Abdullah Wahbeh; Omar El-Gayar; Mohammad Al-Ramahi; Tareq Nasralah; Ahmed Elnoshokaty. “AutoTheme: A Multi-Agent Framework for Inductive Thematic Analysis with LLMs.” Proceedings of the Annual Hawaii International Conference on System Sciences. Year: 2026. Type/status: conference/proceedings paper; peer reviewed. Publication history: peer-reviewed HICSS-59 conference proceedings paper, 2026 DOI: 10.24251/hicss.2026.214. [Official record](#).
- **alghamdi\_2026\_convergence.** AlGhamdi. “From Code Variability to Theme Convergence: Evaluating Large Language Models Against Human Thematic Analysis.” International Journal of Qualitative Methods, 25. Year: 2026. Type/status: journal article; peer reviewed. Publication history: peer-reviewed journal article DOI: 10.1177/16094069261462093. [Official record](#).
- **ta\_0117\_2026\_davis.** Alison H Davis; Gloria Giarratano; Tina Gunaldo. “Comparing Human Insight and AI in Thematic Analysis of Nursing Education Reflections.” The Journal of nursing education. Year: 2026. Type/status: journal article; peer reviewed. Publication history: peer-reviewed Journal of Nursing Education article, published online 2025-08-25 and in volume 65(1), January 2026, pages 55–58 DOI: 10.3928/01484834-20250414-01. [Official record](#).
- **ganesh\_2024\_prompting\_panacea.** Ananya Ganesh; Chelsea Chandler; Sidney D'Mello; Martha Palmer; Katharina von der Wense. “Prompting as Panacea? A Case Study of In-Context Learning Performance for Qualitative Coding of Classroom Dialog.” Proceedings of the 17th International Conference on Educational Data Mining (EDM 2024). Year: 2024. Type/status: peer-reviewed conference paper/poster; peer reviewed. Publication history: Official EDM proceedings paper, pages 835–843; the PDF's footer abbreviates the final author as K. Kann while the title page and

official proceedings page use Katharina von der Wense. DOI: 10.5281/zenodo.12729966. [Official record.](#)

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- **crocker\_2025\_extension\_feedback.** Andrew B. Crocker; Marcelo Schmidt; Juan Diego Tejeda; Howard Rodriguez-Mori. “Proof of Concept: Leveraging Large Language Models for Qualitative Analysis of Participant Feedback.” Journal of Extension 63(1), Article 16. Year: 2025. Type/status: journal article; peer reviewed. Publication history: published in Volume 63, Issue 1 (2025); CC BY-NC-SA 4.0 DOI: 10.34068/joe.63.01.16. [Official record.](#)
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- **ta\_0419\_2026\_wang.** Qile Wang; Prerana Khatiwada; Avinash Chouhan; Ashrey Mahesh; Joy Mwaria; Duy Duc Tran; Kenneth E. Barner; Matthew Louis Mauriello. ““The explanation makes sense”: An Empirical Study on LLM Performance in News Classification and its Influence on Judgment in Human-AI Collaborative Annotation.” arXiv. Year: 2026. Type/status: preprint; not peer reviewed or peer review not established as of 2026-08-24. Publication history: arXiv preprint 2602.19690v1, submitted 23 February 2026; peer review not established DOI: not reported. [Official record.](#)
- **ta\_0373\_2026\_maerz.** Seraphine F. Maerz; Kenneth Benoit. “quallmer: Qualitative Analysis with Large Language Models.” CRAN: Contributed Packages. Year: 2026. Type/status: software/package documentation; not peer reviewed as a research study. Publication history: CRAN software package manual for quallmer version 0.4.0, published 6 May 2026; not a peer-reviewed study DOI: 10.32614/cran.package.quallmer. [Official record.](#)
- **position\_clinical\_ta\_2025.** Seungjun Yi; Joakim Nguyen; Terence Lim; Andrew Well; Joseph Skrovan; Mehak Beri; YongGeon Lee; Kavita Radhakrishnan; Liu Leqi; Mia Markey; Ying Ding. “Position: Thematic Analysis of Unstructured Clinical Transcripts with Large Language Models.” arXiv:2509.14597v2. Year: 2025. Type/status: position paper; preprint/manuscript; peer review not verified. Publication history: arXiv/workshop manuscript; archival proceedings status not verified ; arXiv/workshop position paper; peer-review status not reported DOI: 10.48550/arxiv.2509.14597. [Official record.](#)
- **kapania\_2025\_simulacrum\_stories.** Shivani Kapania; William Agnew; Motahhare Eslami; Hoda Heidari; Sarah E. Fox. “Simulacrum of Stories: Examining Large Language Models as Qualitative Research Participants.” Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. Year: 2025. Type/status: conference paper; peer reviewed. Publication history: CHI 2025 paper; author manuscript also available as arXiv:2409.19430 DOI: 10.1145/3706598.3713220. [Official record.](#)
- **fc\_06\_2026\_han.** Songhee Han; Jueun Shin; Jiyeon Han; Bung-Woo Jun; Hilal Ayan Karabatman. “How Trustworthy Are LLM-as-Judge Ratings for Interpretive Responses? Implications for Qualitative Research Workflows.” arXiv. Year: 2026. Type/status: preprint; not peer reviewed or peer review not established as of 2026-08-24. Publication history: arXiv preprint 2604.00008; no peer-reviewed acceptance or venue was established by 24 August 2026 DOI: 10.48550/arxiv.2604.00008. [Official record.](#)
- **fc\_13\_2026\_lukito.** Steve Lukito; Lifang Li; Susie Chandler; Myrofora Kakoulidou; Georgia Pavlopoulou; Maciej Matejko; Isabel Jackson; Beta Balwani; Tiegna Boyens; Dorian Poulton; Luke Harvey-Nguyen; Amber Johnson; Daniel Stahl; Angus Roberts; Edmund J. S. Sonuga-Barke; the RE-

- STAR Team. "Comparing qualitative thematic analysis and machine-based topic modelling in the analysis of autistic and ADHD young people's accounts of emotions." *Scientific Reports*, volume 16, article 1579. Year: 2026. Type/status: peer-reviewed open-access journal research article; peer reviewed. Publication history: Received 15 August 2025; accepted 30 December 2025; collection/publication 2026; CC BY 4.0. DOI: 10.1038/s41598-025-34570-7. [Official record](#).
- **ta\_0454\_2026\_leonard**. Tammy Leonard; Jim P Stimpson; Miguel Ángel Cano; Wenqi Shi; Ada Abaragu; Song Zhang. "Use of an AI-Based Tool (Human Experiences and Reflections Data Connector) to Improve Discovery and Reuse of Archived Qualitative Data: Protocol for an Algorithm Development and Validation Study." *JMIR research protocols*. Year: 2026. Type/status: peer-reviewed research protocol; peer reviewed. Publication history: peer-reviewed JMIR Research Protocols article, volume 15, e82725, published 2026; protocol outcomes not yet available by 24 August 2026 DOI: 10.2196/82725. [Official record](#).
  - **bertscore\_2020**. Tianyi Zhang; Varsha Kishore; Felix Wu; Kilian Q. Weinberger; Yoav Artzi. "BERTScore: Evaluating Text Generation with BERT." *ICLR 2020*. Year: 2020. Type/status: evaluation-method paper; not reported. Publication history: evaluation-method paper; publication history beyond the linked record not reported DOI: not reported. [Official record](#).
  - **zhou\_et\_al\_2025\_llm\_qualitative\_nursing\_scoping\_review**. Tingting Zhou; Mengyao Xing; Yue Hu; Jun Liang; Qi Ren; Yeqin Yang; Lei Ye. "The Application of Large Language Models in Qualitative Nursing Research: A Scoping Review." *Nursing Outlook* 73(6): 102544. Year: 2025. Type/status: peer-reviewed scoping review; peer reviewed. Publication history: November–December 2025 issue; publisher full HTML inspected. DOI: 10.1016/j.outlook.2025.102544. [Official record](#).
  - **van\_aken\_2022\_behavioral**. Van Aken et al. "What Do You See in this Patient? Behavioral Testing of Clinical NLP Models." *ClinicalNLP 2022*. Year: 2022. Type/status: not reported; not reported. Publication history: not reported; publication history beyond the linked record not reported DOI: 10.18653/v1/2022.clinicalnlp-1.7. [Official record](#).
  - **red\_faced\_rouge\_2019**. Wenyi Tay; Aditya Joshi; Xiuzhen Zhang; Sarvnaz Karimi; Stephen Wan. "Red-faced ROUGE: Examining the Suitability of ROUGE for Opinion Summary Evaluation." *Australasian Language Technology Association Workshop 2019*. Year: 2019. Type/status: evaluation-method paper; not reported. Publication history: evaluation-method paper; publication history beyond the linked record not reported DOI: not reported. [Official record](#).

- **yu\_2025\_same\_company.** Yu D, Liu Z, He H. “Same Company, Same Signal: The Role of Identity in Earnings Call Transcripts.” Findings of ACL. Year: 2025. Type/status: not reported; peer reviewed. Publication history: peer reviewed DOI: 10.18653/v1/2025.findings-acl.946. [Official record](#).
- **zhao\_2021\_calibrate.** Zhao et al. “Calibrate Before Use: Improving Few-Shot Performance of Language Models.” ICML 2021. Year: 2021. Type/status: not reported; not reported. Publication history: not reported; publication history beyond the linked record not reported DOI: not reported. [Official record](#).

### 18.3 Additional cited publications outside the evidence master (3)

- **DeTAILS expanded preprint.** Ansh Sharma; Karen Cochrane; James R. Wallace. “DeTAILS: Deep Thematic Analysis with Iterative LLM Support.” arXiv:2510.17575v2 (2025). Materially expanded, non-peer-reviewed companion to the seven-page CUI paper; counted with that family, not as a second evidence identity. [Official preprint](#).
- **Kempny scoping review.** Christian Kempny; Julian Frings; Paul Rust; Sven Meister; Leonard Fehring. “The Use and Methodological Reporting of Large Language Models in Qualitative Research: A Scoping Review.” *BMC Medical Research Methodology* 26, 137 (2026). Peer-reviewed secondary review; TA-0428 is fully inspected and recorded in the canonical flow as an evidence-map exclusion, then cited contextually here outside the 237-row master. DOI: 10.1186/s12874-026-02913-1. [Official record](#).
- **MoverScore.** Wei Zhao; Maxime Peyrard; Fei Liu; Yang Gao; Christian M. Meyer; Steffen Eger. “MoverScore: Text Generation Evaluating with Contextualized Embeddings and Earth Mover Distance.” *EMNLP-IJCNLP 2019*. Peer-reviewed proceedings paper. DOI: 10.18653/v1/D19-1053. [Official record](#).

### 18.4 Cited publication-status and reproducibility artifacts (12)

- **Auto-TA status record.** OpenReview forum with exact venue/decision label [ACL - SRW 2025 Poster](#); it does not establish an archival ACL Anthology publication. [OpenReview](#).
- **SFT-TA status record.** OpenReview forum with exact venue/decision label [ML4H 2025 Reject from Proceedings as Findings Poster](#); it does not establish an archival PMLR publication. [OpenReview](#).
- **Dai code repository.** Linked implementation artifact for the official 2023 Findings paper. [Repository](#).

- **LLooM code repository.** Linked implementation artifact for the official CHI 2024 paper. [Repository.](#)
- **HiCode code repository.** Linked implementation artifact for the official EMNLP 2025 paper. [Repository.](#)
- **DeTAILS code repository.** Linked implementation artifact for the official CUI 2025 paper. [Repository.](#)
- **CentaurTA code repository.** Linked implementation artifact for the official Findings of ACL 2026 paper. [Repository.](#)
- **TAMA code repository.** Linked implementation artifact for the official ACM article. [Repository.](#)
- **LLM-TA code repository.** Linked implementation artifact for the official preprint. [Repository.](#)
- **Muse evaluation repository.** Linked evaluation-data artifact for the official ACL Industry paper. [Repository.](#)
- **Trauma-study code repository.** Linked artifact for the official Findings of ACL 2026 paper. [Repository.](#)
- **Childbirth/parenthood repository.** Linked code/prompt artifact for the official ICWSM paper; the review flags the printed API key and paper-code inconsistencies. [Repository.](#)

## 19. Appendices: search strings, screening, extraction, and quality rubric

### 19.1 Exact database queries, dates, filters, and counts

All auditable database queries were run on **24 August 2026**. The table reproduces the stored query text; record identifiers and complete source responses remain in the named JSON exports. Crossref's reported multi-million totals are fuzzy relevance estimates and were not treated as screenable counts.

| Source/query ID     | Exact stored query                                                                                                                                                                                                                                                           | Filters                                                                               | Retrieved/retained |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------|
| PubMed<br>PUBMED_Q1 | <code>(( "large language model"[Title/Abstract] OR LLM[Title/Abstract] OR "generative AI"[Title/Abstract]) AND ("thematic analysis"[Title/Abstract] OR "theme generation"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication])</code> | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 229                |
| PubMed<br>PUBMED_Q2 | <code>(( "large language model"[Title/Abstract] OR LLM[Title/Abstract]) AND ("qualitative coding"[Title/Abstract] OR "inductive coding"[Title/Abstract] OR "open coding"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication])</code>  | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 8                  |
| PubMed<br>PUBMED_Q3 | <code>(( "large language model"[Title/Abstract] OR LLM[Title/Abstract]) AND ("qualitative content analysis"[Title/Abstract] OR "qualitative data analysis"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication])</code>                | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 25                 |

| Source/query ID     | Exact stored query                                                                                                                                                                                                                                                        | Filters                                                                               | Retrieved/retained |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------|
| PubMed<br>PUBMED_Q4 | (( "human-AI collaboration"[Title/Abstract] OR "human-in-the-loop"[Title/Abstract]) AND ("thematic analysis"[Title/Abstract] OR "qualitative research"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication])                        | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 47                 |
| PubMed<br>PUBMED_Q5 | (( "multi-agent"[Title/Abstract] OR "LLM agent"[Title/Abstract]) AND ("thematic analysis"[Title/Abstract] OR "qualitative coding"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication])                                             | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 2                  |
| PubMed<br>PUBMED_Q6 | (( "codebook induction"[Title/Abstract] OR "codebook generation"[Title/Abstract] OR "codebook refinement"[Title/Abstract]) AND (LLM[Title/Abstract] OR "large language model"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication]) | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 0                  |
| PubMed<br>PUBMED_Q7 | (( "clinical transcript"[Title/Abstract] OR "patient interview"[Title/Abstract]) AND ("large language model"[Title/Abstract] OR LLM[Title/Abstract] OR "automated coding"[Title/Abstract])) AND ("2020/01/01"[Date - Publication] : "2026/08/24"[Date - Publication])     | Publication date 2020-01-01 through 2026-08-24;<br>title/abstract fields as specified | 1                  |

| Source/query ID | Exact stored query                                                                                                                                                        | Filters                                                                      | Retrieved/retained |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------|
| arXiv ARXIV_Q1  | (all:LLM AND all:"thematic analysis") AND submittedDate: [202001010000 TO 202608242359]                                                                                   | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records | 116                |
| arXiv ARXIV_Q2  | (all:LLM AND (all:"qualitative coding" OR all:"inductive coding" OR all:"open coding")) AND submittedDate: [202001010000 TO 202608242359]                                 | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records | 77                 |
| arXiv ARXIV_Q3  | (all:LLM AND (all:"qualitative content analysis" OR all:"qualitative data analysis")) AND submittedDate: [202001010000 TO 202608242359]                                   | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records | 22                 |
| arXiv ARXIV_Q4  | ((all:"human-AI collaboration" OR all:"human-in-the-loop") AND (all:"thematic analysis" OR all:"qualitative research")) AND submittedDate: [202001010000 TO 202608242359] | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records | 29                 |
| arXiv ARXIV_Q5  | ((all:"multi-agent" OR all:"LLM agent") AND (all:"thematic analysis" OR all:"qualitative coding")) AND submittedDate: [202001010000 TO 202608242359]                      | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records | 12                 |
| arXiv ARXIV_Q6  | (all:LLM AND (all:"codebook induction" OR all:"codebook generation" OR all:"codebook refinement")) AND submittedDate: [202001010000 TO 202608242359]                      | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records | 6                  |

| Source/query ID      | Exact stored query                                                                                                                                                | Filters                                                                                                                      | Retrieved/retained |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------|
| arXiv ARXIV_Q7       | (all:LLM AND (all:"clinical transcript" OR all:"patient interview")) AND submittedDate: [202001010000 TO 202608242359]                                            | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records                                                 | 14                 |
| arXiv ARXIV_Q8       | ((all:"identity leakage" OR all:"speaker identity" OR all:"shortcut learning") AND all:text) AND submittedDate: [202001010000 TO 202608242359]                    | Submitted 2020-01-01 through 2026-08-24; relevance sort; maximum 200 records                                                 | 200                |
| Crossref CROSSREF_Q1 | query.title: large language model thematic analysis theme generation                                                                                              | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| Crossref CROSSREF_Q2 | query.title: large language model qualitative coding inductive coding open coding                                                                                 | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| Crossref CROSSREF_Q3 | query.title: large language model qualitative content analysis qualitative data analysis                                                                          | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| Crossref CROSSREF_Q4 | query.title: human AI collaboration human in the loop thematic analysis qualitative research                                                                      | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| Crossref CROSSREF_Q5 | query.title: multi-agent LLM agent thematic analysis qualitative coding                                                                                           | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| Crossref CROSSREF_Q6 | query.title: codebook induction codebook generation codebook refinement LLM                                                                                       | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| Crossref CROSSREF_Q7 | query.title: clinical transcript patient interview large language model automated coding                                                                          | from-pub-date:2020-01-01, until-pub-date:2026-08-24; relevance sort; rows=100                                                | 100                |
| ACL Anthology ACL_Q1 | TITLE contains any of ['large language model', 'llm', 'generative ai', 'generative artificial intelligence'] AND any of ['thematic analysis', 'theme generation'] | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching | 1                  |

| Source/query ID                    | Exact stored query                                                                                                                                                                                    | Filters                                                                                                                                           | Retrieved/retained          |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| ACL Anthology<br>ACL_Q2            | TITLE contains any of ['large language model', 'llm'] AND any of ['qualitative coding', 'inductive coding', 'open coding']                                                                            | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 3                           |
| ACL Anthology<br>ACL_Q3            | TITLE contains any of ['large language model', 'llm'] AND any of ['qualitative content analysis', 'qualitative data analysis']                                                                        | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 4                           |
| ACL Anthology<br>ACL_Q4            | TITLE contains any of ['human-ai collaboration', 'human ai collaboration', 'human-in-the-loop', 'human in the loop'] AND any of ['thematic analysis', 'qualitative research', 'qualitative analysis'] | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 0                           |
| ACL Anthology<br>ACL_Q5            | TITLE contains any of ['multi-agent', 'multi agent', 'llm agent'] AND any of ['thematic analysis', 'qualitative coding']                                                                              | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 0                           |
| ACL Anthology<br>ACL_Q6            | TITLE contains any of ['llm', 'large language model'] AND any of ['codebook induction', 'codebook generation', 'codebook refinement']                                                                 | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 0                           |
| ACL Anthology<br>ACL_Q7            | TITLE contains any of ['llm', 'large language model', 'automated coding'] AND any of ['clinical transcript', 'patient interview']                                                                     | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 0                           |
| ACL Anthology<br>ACL_Q8_SUPPLEMENT | TITLE contains any of [] AND any of ['thematic analysis', 'qualitative coding', 'qualitative data analysis', 'inductive coding', 'open codes']                                                        | article/inproceedings records in ACL Anthology snapshot generated 2026-08-24; title field; case-insensitive literal matching                      | 12                          |
| OpenReview<br>OPENREVIEW_Q1        | "thematic analysis"                                                                                                                                                                                   | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 379 retrieved / 78 retained |

| Source/query ID                             | Exact stored query             | Filters                                                                                                                                           | Retrieved/retained         |
|---------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| OpenReview<br><a href="#">OPENREVIEW_Q2</a> | "qualitative coding"           | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 66 retrieved / 12 retained |
| OpenReview<br><a href="#">OPENREVIEW_Q3</a> | "inductive coding"             | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 33 retrieved / 16 retained |
| OpenReview<br><a href="#">OPENREVIEW_Q4</a> | "open coding"                  | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 55 retrieved / 13 retained |
| OpenReview<br><a href="#">OPENREVIEW_Q5</a> | "qualitative data analysis"    | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 39 retrieved / 10 retained |
| OpenReview<br><a href="#">OPENREVIEW_Q6</a> | "qualitative content analysis" | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 52 retrieved / 6 retained  |
| OpenReview<br><a href="#">OPENREVIEW_Q7</a> | "codebook generation"          | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 52 retrieved / 1 retained  |
| OpenReview<br><a href="#">OPENREVIEW_Q8</a> | "codebook refinement"          | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 5 retrieved / 1 retained   |
| OpenReview<br><a href="#">OPENREVIEW_Q9</a> | "clinical transcript"          | phrase search; first 1,000 relevance-ranked results; created by 2026-08-24; client filter requires an LLM term and qualitative-analysis task term | 2 retrieved / 0 retained   |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

Targeted ACM DL, IEEE Xplore, Google Scholar, and Semantic Scholar searches used exact seed titles, authors, DOIs, venue names, citing-paper queries, and same-family searches; the interfaces did not

expose stable auditable bulk totals. Web of Science and Scopus were inaccessible. Citation chaining queried fifteen supplied/prespecified seeds through available indexes, retained parent-seed/source provenance, repeated an OpenAlex confirmation pass, and required inspected full text before inclusion. This focused procedure does not support a citation-saturation claim.

## 19.2 Screening, retrieval, exclusion, and publication-family logs

### Identity-deduplicated PRISMA arithmetic

The original source exports contained 1,556 records. Exact normalized DOI/title deduplication removed 196, leaving 1,360 database identities. Initial deterministic triage (122 high, 612 uncertain, 626 low) set review priority only. Terminal screening excluded 1,116 database records at title/abstract and advanced 244. The primary unique-identity full-text flow is  $317 = 53 + 264$ , and the assessed/resolved flow is  $264 = 237 + 18 + 9$ . The 237 retained identities comprise 181 core and 56 noncore records.

| Unique identity stratum                                                             | Sought     | Not retrieved | Assessed/resolved | Retained   | Full-text excluded | Family resolution |
|-------------------------------------------------------------------------------------|------------|---------------|-------------------|------------|--------------------|-------------------|
| database/register union                                                             | 244        | 44            | 200               | 174        | 17                 | 9                 |
| focused forward citation chaining outside the database union                        | 34         | 8             | 26                | 25         | 1                  | 0                 |
| backward citation chaining                                                          | 3          | 1             | 2                 | 2          | 0                  | 0                 |
| other unique prespecified, supplied, targeted, or methodological-foundation reports | 36         | 0             | 36                | 36         | 0                  | 0                 |
| <b>All unique identities</b>                                                        | <b>317</b> | <b>53</b>     | <b>264</b>        | <b>237</b> | <b>18</b>          | <b>9</b>          |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

Focused forward chaining produced 59 route records, of which 21 overlapped the database union. The outside-union identity flow was 38 screened → four title/abstract exclusions → 34 sought → eight not

retrieved +26 assessed →25 retained (16 core, nine adjacent) + one full-text exclusion. The following route-event table is secondary and nonadditive because it preserves those overlaps.

| Secondary route event                                                     | Sought     | Not retrieved | Assessed/ resolved | Retained   | Full-text excluded | Family resolution |
|---------------------------------------------------------------------------|------------|---------------|--------------------|------------|--------------------|-------------------|
| database/<br>register union                                               | 244        | 44            | 200                | 174        | 17                 | 9                 |
| focused<br>forward<br>citation<br>chaining                                | 54         | 10            | 44                 | 41         | 1                  | 2                 |
| backward<br>citation<br>chaining                                          | 3          | 1             | 2                  | 2          | 0                  | 0                 |
| other<br>prespecified,<br>supplied, or<br>targeted full-<br>text evidence | 20         | 0             | 20                 | 20         | 0                  | 0                 |
| <b>All route<br/>events</b>                                               | <b>321</b> | <b>55</b>     | <b>266</b>         | <b>237</b> | <b>18</b>          | <b>11</b>         |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

### Full-text exclusions (18)

Every full-text exclusion has an inspected-record-specific reason. The 17 substantive ineligibility exclusions contribute no methods or results to the synthesis. TA-0428 is the distinct, explicitly disclosed 18th evidence-map boundary exclusion: that fully inspected secondary review contributes only contextual reporting-practice statistics and remains outside the 237 retained evidence identities and all quality scoring.

| Route/ID                                             | Title                                                                                                                                                    | Source       | Concise full-text reason                                                                                                                                                                                                    |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register<br>union / <a href="#">TA-0008</a> | A Data-Driven Analysis for Engineering Conferences: The Institute of Industrial and Systems Engineering (IISE) Annual Conference Proceedings (2002-2025) | not reported | The LLM maps conference-paper titles to predetermined domains and subareas for bibliometric cartography; the word 'thematic' denotes topic/domain classification, not qualitative coding or interpretive theme development. |

| Route/ID                                          | Title                                                                                                                                 | Source              | Concise full-text reason                                                                                                                                                                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-0047</a> | An Evaluation of Chat Safety Moderations in Roblox                                                                                    | not reported        | The LLM is evaluated and used as a safety-message classifier/prefilter; humans, not the LLM, conduct iterative open and axial qualitative coding to saturation. It is therefore a substantive safety study rather than a computational qualitative-analysis method/evaluation.               |
| database/register union / <a href="#">TA-0428</a> | The use and methodological reporting of large language models in qualitative research: a scoping review.                              | <a href="#">DOI</a> | Peer-reviewed secondary scoping review inspected and cited for contextual reporting-practice statistics, but kept outside the 237-row primary/adjacent evidence master; it is therefore an assessed evidence-map exclusion rather than a retained evidence identity.                         |
| database/register union / <a href="#">TA-0429</a> | Thematic Analysis of Accepted Exception Requests for Heart Transplant Candidates Using a Large Language Model                         | <a href="#">DOI</a> | The only published item is a one-page meeting abstract rather than a full research report. It cannot support the review's required full-text methodological extraction and quality appraisal.                                                                                                |
| database/register union / <a href="#">TA-0664</a> | Generative AI for Thematic Analysis in a Maternal Health Study: Coding Semi-structured Interviews using Large Language Models (LLMs). | <a href="#">DOI</a> | Full preprint was inspected and is substantively the same publication family as TA-0665; medRxiv v2 DOI 10.1101/2024.09.16.24313707 is superseded by the peer-reviewed version of record and must not be counted twice.                                                                      |
| database/register union / <a href="#">TA-0696</a> | Integrating Qualitative Comparative Analysis With Reflexive Thematic Analysis in Theme Development                                    | <a href="#">DOI</a> | Full-text inspection establishes that 'QCA' means qualitative comparative analysis and that the article integrates that human analytic method with reflexive thematic analysis. It contains no AI, LLM, computational analyst, or directly applicable computational-evaluation contribution. |

| Route/ID                                          | Title                                                                                                                                                 | Source              | Concise full-text reason                                                                                                                                                                                    |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-0921</a> | Clinician voices on ethics of LLM integration in healthcare: a thematic analysis of ethical concerns and implications.                                | <a href="#">DOI</a> | The complete article uses human thematic analysis of clinician interviews to study ethical concerns about LLM adoption; an LLM is the study topic, not the qualitative analyst.                             |
| database/register union / <a href="#">TA-0929</a> | Concept Over Time Analysis: Unveiling Temporal Patterns for Qualitative Data Analysis                                                                 | <a href="#">DOI</a> | The complete ACL demo paper classifies sentences against researcher-specified concepts and visualizes their prevalence over time; it does not generate or evaluate qualitative codes or themes with an LLM. |
| database/register union / <a href="#">TA-0965</a> | Harmful Speech Detection by Language Models Exhibits Gender-Queer Dialect Bias                                                                        | not reported        | The complete paper audits dialect and gender bias in harmful-speech classifiers; it does not perform or evaluate qualitative coding or thematic interpretation.                                             |
| database/register union / <a href="#">TA-0995</a> | Moving Beyond LDA: A Comparison of Unsupervised Topic Modelling Techniques for Qualitative Data Analysis of Online Communities                        | not reported        | The complete paper compares LDA, NMF, and BERTopic for corpus topic discovery and researcher usability, without an LLM-based qualitative coding or theme-generation method.                                 |
| database/register union / <a href="#">TA-1003</a> | Navigating the Shortcut Maze: A Comprehensive Analysis of Shortcut Learning in Text Classification by Language Models                                 | not reported        | The complete paper is a shortcut-learning benchmark for supervised text classification; its validity findings are general NLP evidence rather than a qualitative-analysis study.                            |
| database/register union / <a href="#">TA-1036</a> | The African Woman is Rhythmic and Soulful: An Investigation of Implicit Biases in LLM Open-ended Text Generation                                      | not reported        | The complete paper audits cultural and gender stereotypes in open-ended LLM generation; it does not use an LLM to code or interpret qualitative data.                                                       |
| database/register union / <a href="#">TA-1095</a> | Improved prompting and process for writing user personas with LLMs, using qualitative interviews: Capturing behaviour and personality traits of users | not reported        | The complete paper evaluates LLM generation of user personas from interview-derived material; persona authoring is a downstream design task, not a study of qualitative coding or theme induction.          |

| Route/ID                                                                          | Title                                                                                                                  | Source                          | Concise full-text reason                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1145</a>                                 | Writing user personas with Large Language Models: Testing phase 6 of a Thematic Analysis of semi-structured interviews | not reported                    | The complete paper tests phase-six persona writing after thematic-analysis inputs have already been produced; it does not evaluate the LLM as the qualitative coder or theme analyst.                                                                                                                     |
| database/register union / <a href="#">TA-1317</a>                                 | Landscape of Large Language Models in Global English News: Topics, Sentiments, and Spatiotemporal Analysis             | not reported                    | The preferred ICWSM full text uses BERTopic, lexicon sentiment analysis, and a human-built codebook to characterize news coverage; it does not evaluate an LLM-supported qualitative-analysis workflow.                                                                                                   |
| database/register union / <a href="#">TA-1339</a>                                 | Setting the AI Agenda - Evidence from Sweden in the ChatGPT Era                                                        | not reported                    | The complete 2024 manuscript uses three human coders for qualitative content analysis of 74 Swedish opinion articles and explicitly rejects an LLM approach after a brief Gemma experiment; ChatGPT is the debate topic, not the analyst.                                                                 |
| database/register union / <a href="#">TA-1354</a>                                 | Unmasking Public Sentiment: A Sample Efficient Approach to Analyzing Twitter Opinion on US Aid to Ukraine              | not reported                    | HICSS 2025, DOI 10.24251/HICSS.2025.300; LLM is only a sentiment classifier and themes are manual                                                                                                                                                                                                         |
| focused forward citation chaining / <a href="#">doi:10.48550/arxiv.2603.01930</a> | From Variance to Invariance: Qualitative Content Analysis for Narrative Graph Annotation                               | <a href="#">official record</a> | The operational study uses human qualitative content analysis to construct a narrative-graph annotation scheme and graph metrics. LLM references and a GLiNER pre-annotation component do not constitute LLM-assisted coding, theme development, or qualitative interpretation in the evaluated workflow. |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

### Reports not retrieved after documented attempts (53)

These plausible or contextual records remain outside the evidence master. Abstracts and publisher metadata were used only for the retrieval disposition, never for methodological or numerical synthesis.

| Route/ID                          | Title                                                                                                                                                                   | Source              | Retrieval disposition                                                                                                                                                                                                                                                              |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / TA-0027 | Advancing qualitative analysis in nursing research: Comparing text analysis rigor and efficiency of large language models with human-coded analysis for interview data. | <a href="#">DOI</a> | Title and abstract make this a clearly core comparative LLM-versus-human qualitative-analysis study, but publisher and API attempts did not expose inspectable full text. No methodological extraction or quality score is created from the abstract.                              |
| database/register union / TA-0037 | AI-Assisted Qualitative Data Analysis                                                                                                                                   | <a href="#">DOI</a> | The abstract/metadata describe a contextual methodological chapter on AI-assisted qualitative data analysis, but only a publisher sample/metadata was available. No full-text-grounded evidence row is created.                                                                    |
| database/register union / TA-0059 | Artificial intelligence and large language model-powered chatbots to support qualitative content analysis                                                               | <a href="#">DOI</a> | The chapter is directly relevant as methodological/contextual guidance for LLM-supported qualitative content analysis, but only the official abstract and book metadata were accessible. No extraction or quality score is inferred from them.                                     |
| database/register union / TA-0065 | Artificial Intelligence in Systematic Literature Reviews: Social Work Ethics, Application, and Feasibility.                                                             | <a href="#">DOI</a> | The abstract indicates potentially eligible LLM-assisted qualitative work inside a systematic-review case study, but the publisher article was paywalled and the institutional repository exposed metadata without a file. Full-text eligibility and extraction remain unresolved. |
| database/register union / TA-0133 | Data sharing statements in five high-impact anesthesiology journals (2020-2023): a cross-sectional study.                                                               | <a href="#">DOI</a> | The official preview confirms GPT-4-assisted exploratory thematic grouping with two-investigator manual refinement/application, making the study plausibly core, but the methods/results body is subscription-only. No evidence row or score is created without full text.         |

| Route/ID                          | Title                                                                                                                                      | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                 |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / TA-0140 | Demonstration of Generative AI Assisted Inductive Coding: Exploring Sociotechnical Imaginaries for Qualitative Research                    | <a href="#">DOI</a> | The title/abstract clearly describe a core five-step human-in-the-loop inductive-coding demonstration, but the otherwise open AISEL PDF endpoint repeatedly returned 403/failed fetch. No methodological claims are extracted from the abstract.                                                      |
| database/register union / TA-0251 | Integrating Large Language Models into Thematic Analysis Workflows for Healthcare Research.                                                | <a href="#">DOI</a> | The two-page proceedings paper is directly about an LLM/human thematic-analysis workflow, but the IOS Press article/PDF was not accessible beyond metadata and abstract. No methods are reconstructed from its abstract.                                                                              |
| database/register union / TA-0276 | Large Language Model-Based Classification of Social-Justice Advocacy Competencies: Concordance with Human Qualitative Coding               | <a href="#">DOI</a> | The publisher PDF was retrieved, but its text layer is incomplete/garbled and predominantly Korean; the available artifact did not support reliable line-level verification of methods, results, or limitations. No master-ready extraction is made.                                                  |
| database/register union / TA-0318 | More than a gut feeling: large language model thematic analysis of social media posts from inflammatory bowel disease patient communities. | <a href="#">DOI</a> | The abstract is clearly in scope (GPT-4 thematic analysis of 104,123 Reddit posts with NLP and human-chart-review validation), but the publisher full text remained access-controlled and no lawful open manuscript was located. Methodological claims are therefore not extracted from the abstract. |

| Route/ID                          | Title                                                                                                                              | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / TA-0346 | Patient voice at scale: artificial intelligence-assisted qualitative analysis of patient-clinician rapport in public social media. | <a href="#">DOI</a> | The abstract is clearly in scope (human-led reflexive thematic analysis with LLM clustering suggestions and provisional summaries), but the publisher's nominally free article could not be retrieved because its delivery endpoint returned an access challenge; an automated PMC retrieval was a title mismatch and was quarantined. No methodological extraction is made from the abstract. |
| database/register union / TA-0560 | Automating Thematic analysis of COVID-19 Vaccine Hesitancy in Alaska with multi-agent LLM systems                                  | <a href="#">DOI</a> | The abstract describes multi-agent LLM thematic analysis of 87 semi-structured interviews, with agents representing four disciplinary perspectives and producing five themes. This is plausibly core, but the TechRxiv DOI and manuscript endpoint did not yield a reproducibly inspectable full text, so no methodological or performance claims beyond the abstract were extracted.          |
| database/register union / TA-0577 | Can Large Language Models Perform Inductive Qualitative Coding? - Evaluating GPT-4 on Korean-Language Texts                        | <a href="#">DOI</a> | The official metadata abstract reports GPT-4 inductive coding of 497 Korean documentary quotations, including BERTScore and character-level agreement results. This is plausibly core and uniquely informative for non-English evaluation, but the journal/KISS full text was access-restricted; abstract-only results were not promoted to a 52-field extraction or quality score.            |

| Route/ID                          | Title                                                                                                                                         | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / TA-0640 | Evaluating LLM Assisted Qualitative Analysis in Medical Education Research: A Comparison of Human and AI-Generated Thematic Coding (Preprint) | <a href="#">DOI</a> | The official preprint abstract reports GPT-4o coding of a four-participant medical-student focus group and separate inductive and deductive comparisons, including code-level agreement and kappa. It is plausibly core, but the official preprint manuscript was not reproducibly retrievable; no abstract-only extraction or quality score was created.                                  |
| database/register union / TA-0684 | Human-AI Collaboration Pipeline for Thematic Analysis: Example of VR-Supported Pulmonary Rehabilitation                                       | <a href="#">DOI</a> | Official metadata describes a three-agent thematic-analysis pipeline applied to interviews from nine VR pulmonary-rehabilitation participants and reports a Sentence-T5-XXL similarity comparison. This is plausibly core, but the exact 2025 IEEE full text was not reproducibly available. A related 2026 report was not treated as proof that this conference paper had been inspected. |
| database/register union / TA-0726 | Large Language Model Versus Human-Generated Thematic Analysis in Otolaryngology Qualitative Research.                                         | <a href="#">DOI</a> | The title and indexed metadata explicitly describe a human-versus-LLM thematic-analysis comparison, so the paper remains plausibly core. The short Laryngoscope article was not available through a reproducibly inspectable publisher, repository, or indexed full-text endpoint; no methodological claims were inferred from its title.                                                  |

| Route/ID                          | Title                                                                                            | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------|--------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / TA-0891 | A New Method Supporting Qualitative Data Analysis Through Prompt Generation for Inductive Coding | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The title directly names open/inductive coding, qualitative-analysis workflow, but the union has no abstract; full text is required before eligibility can be decided.                             |
| database/register union / TA-0915 | Can a zero-shot learning Large Language Model code complex interview data?                       | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 5 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; openalex_resolved_oa_pdf; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to open/inductive coding, deductive coding; full text is required before methodological extraction or inclusion. |
| database/register union / TA-0918 | ChatGPT in Thematic Analysis: Can AI become a research assistant in qualitative research?        | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |

| Route/ID                                          | Title                                                                                                             | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-0922</a> | CoAlcoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis        | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to qualitative-analysis workflow; full text is required before methodological extraction or inclusion.                                                               |
| database/register union / <a href="#">TA-0956</a> | Exploring the synergy of human and AI-driven approaches in thematic analysis for qualitative educational research | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 5 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; openalex_resolved_oa_pdf; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |

| Route/ID                                          | Title                                                                                                                             | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-0971</a> | Human-AI Collaboration in Thematic Analysis using ChatGPT: A User Study and Design Recommendations                                | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation; full text is required before methodological extraction or inclusion.                                                                                                                                                                              |
| database/register union / <a href="#">TA-0982</a> | Large Language Models Can Enable Inductive Thematic Analysis of a Social Media Corpus in a Single Prompt: Human Validation Study. | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 10 documented attempts (crossref_metadata_lookup; doi_content_negotiation; europe_pmc_confirmation_lookup; europe_pmc_fulltext_lookup; indexed_landing_or_file; official_europe_pmc_full_text_xml; openalex_oa_metadata_lookup; openalex_resolved_oa_pdf; pubmed_central_id_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation; full text is required before methodological extraction or inclusion. |

| Route/ID                                          | Title                                                                                                                              | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-0985</a> | Leveraging Large Language Models for Automating Inductive Qualitative Coding: A Comparative Study of Prompt Engineering Techniques | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |
| database/register union / <a href="#">TA-1019</a> | Reflexive Content Analysis: An Approach to Qualitative Data Analysis, Reduction, and Description                                   | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The abstract introduces reflexive content analysis and is relevant methodological guidance, but it contains no computational contribution.                     |

| Route/ID                          | Title                                                                             | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|-----------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / TA-1057 | Academic publisher guidelines on AI usage: A ChatGPT supported thematic analysis. | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 10 documented attempts<br>(crossref_metadata_lookup; doi_content_negotiation; europe_pmc_confirmation_lookup; europe_pmc_fulltext_lookup; indexed_landing_or_file; official_europe_pmc_full_text_xml; openalex_oa_metadata_lookup; openalex_resolved_oa_pdf; pubmed_central_id_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation; full text is required before methodological extraction or inclusion. |
| database/register union / TA-1080 | Computational Thematic Analysis of Poetry via Bimodal Large Language Models       | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts<br>(crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The abstract operationalizes 'thematic analysis' as supervised multilabel poem-topic classification, making it useful evidence of methodological conflation but not a core qualitative-analysis study.                                                                                                                                           |

| Route/ID                                          | Title                                                                                                          | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1132</a> | Supporting Qualitative Analysis with Large Language Models: Combining Codebook with GPT-3 for Deductive Coding | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 5 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; openalex_resolved_oa_pdf; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to deductive coding, codebook development, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |
| database/register union / <a href="#">TA-1141</a> | Using Large Language Models to Support Thematic Analysis in Empirical Legal Studies                            | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to open/inductive coding, thematic analysis/theme generation; full text is required before methodological extraction or inclusion.                                       |

| Route/ID                                          | Title                                                                                         | Source              | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1142</a> | Using the Reproducible Open Coding Kit & Epistemic Network Analysis to model qualitative data | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The title links a reproducible open-coding kit with epistemic-network analysis; it is a pre-LLM computational qualitative method requiring full-text assessment.                  |
| database/register union / <a href="#">TA-1236</a> | Word Association Thematic Analysis: Theme Detection                                           | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The title proposes word-association theme detection, a related non-LLM computational method whose interpretive status requires full text.                                         |
| database/register union / <a href="#">TA-1248</a> | Establishing Interrater Reliability in Qualitative Coding                                     | <a href="#">DOI</a> | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 4 documented attempts (crossref_metadata_lookup; doi_content_negotiation; openalex_oa_metadata_lookup; unpaywall_lookup). It is not included. Routing basis: The title is directly relevant methodological guidance on inter-rater reliability in qualitative coding, but the union supplies no abstract and it is not a computational system. |

| Route/ID                                          | Title                                                                                                                                                                   | Source       | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1270</a> | A New Method Using LLMs for Keypoints Generation in Qualitative Data Analysis                                                                                           | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The title directly names qualitative-analysis workflow, but the union has no abstract; full text is required before eligibility can be decided.                                                    |
| database/register union / <a href="#">TA-1275</a> | Analyzing Nobel Prize Literature with Large Language Models                                                                                                             | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The abstract reports LLM analysis of Nobel literature; full text is required to determine whether the output is qualitative interpretation or only topic/label extraction.                         |
| database/register union / <a href="#">TA-1278</a> | Artificial intelligence in qualitative analysis: a practical guide and reflections based on results from using GPT to analyze interview data in a substance use program | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |

| Route/ID                                          | Title                                                                                                                               | Source       | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1292</a> | Do Artificial Intelligence Clients Speak Like Human Clients?<br>Exploring GPT-4's Content-Level Performance in Counseling Role-Play | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts<br>(indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup)<br>. It is not included. Routing basis: The abstract compares GPT-4 and human-client content, which may inform qualitative-output validity but is not clearly a qualitative-analysis method.                         |
| database/register union / <a href="#">TA-1310</a> | GPT Models Can Perform Thematic Analysis in Public Health Studies, Akin to Qualitative Researchers                                  | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts<br>(indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup)<br>. It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation; full text is required before methodological extraction or inclusion.          |
| database/register union / <a href="#">TA-1314</a> | Image Embeddings from Social Media: Computer Vision and Human in the Loop Applications for Social Movement Messaging                | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts<br>(indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup)<br>. It is not included. Routing basis: The abstract describes image embeddings with human-in-the-loop analysis of social-movement messages; it is adjacent multimodal evidence rather than text qualitative coding. |

| Route/ID                                          | Title                                                                                                                                 | Source       | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1319</a> | LLM-as-Peer-Debrief: A Multi-Agent Framework with Multi-Perspective Refinement for Qualitative Data Analysis                          | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |
| database/register union / <a href="#">TA-1329</a> | Position: Bridge Human Interpretation and Machine Representation With Explicit Specification For Qualitative Data Analysis In LLM Era | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to qualitative-analysis workflow; full text is required before methodological extraction or inclusion.                                     |
| database/register union / <a href="#">TA-1331</a> | Post-GPT Policy: Risk and Regulation in EU AI Discourse                                                                               | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to a directly named qualitative-analysis task; full text is required before methodological extraction or inclusion.                        |

| Route/ID                                          | Title                                                                                                                                             | Source       | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1335</a> | QualAlign: Benchmarking Automated Qualitative Coding Against Human Schemas                                                                        | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to open/inductive coding, codebook development, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |
| database/register union / <a href="#">TA-1337</a> | Sentiment Dynamics of Graduating Students-to-Employees Transitions: A Paired Sample LLM-Based Analysis of Expectations and Real-World Experiences | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to thematic analysis/theme generation; full text is required before methodological extraction or inclusion.                                         |
| database/register union / <a href="#">TA-1351</a> | Tracing Coordination Dynamics in Multi-Turn LLM Discussions                                                                                       | not reported | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to open/inductive coding; full text is required before methodological extraction or inclusion.                                                      |

| Route/ID                                                                             | Title                                                                                                                                | Source                          | Retrieval disposition                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1356</a>                                    | Using Large Language Model to Support Flexible and Structural Inductive Qualitative Analysis                                         | not reported                    | The record remained plausibly eligible at title/abstract screening, but full text was not retrieved after 3 documented attempts (indexed_landing_or_file; official_openreview_pdf; openalex_oa_metadata_lookup) . It is not included. Routing basis: The supplied abstract describes an LLM/NLP contribution to codebook development, qualitative-analysis workflow; full text is required before methodological extraction or inclusion. |
| focused forward citation chaining / <a href="#">doi:10.1007/s10515-025-00587-5</a>   | QuaRUM: qualitative data analysis-based retrieval-augmented UML domain model from requirements documents                             | <a href="#">official record</a> | QuaRUM is clearly a high-priority core candidate, but the official Springer page exposes only a subscription preview and no lawful full manuscript was located. The public code repository is an implementation companion, not a substitute for paper-level methods/results; no evidence record is created from the abstract.                                                                                                             |
| focused forward citation chaining / <a href="#">doi:10.1007/978-3-032-29760-0_61</a> | From Examples to Rules? Exploring Inductive Reverse Engineering and Deductive Few-Shot Coding via LLMs for Qualitative Data Analysis | <a href="#">official record</a> | The title and publisher scope identify a likely core qualitative-coding experiment, but the Springer page provided only a subscription preview and no lawful author manuscript was located. No methodological or numerical evidence is extracted from the abstract.                                                                                                                                                                       |
| focused forward citation chaining / <a href="#">doi:10.1177/10451595261440472</a>    | Fostering Critical Engagement With Generative AI in Qualitative Research Through Andragogy                                           | <a href="#">official record</a> | Title/abstract evidence signals an adjacent conceptual-andragogy paper about critical GenAI engagement in graduate qualitative-research teaching rather than a demonstrated primary QDA evaluation, but the full article was not retrievable. The record remains an explicit nonretrieval rather than an abstract-grounded evidence row.                                                                                                  |

| Route/ID                                                              | Title                                                                                                      | Source                          | Retrieval disposition                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| focused forward citation chaining / doi:10.1080/10494820.2026.2617988 | CATALYST: learner feedback mining with an LLM-based automated thematic analysis                            | <a href="#">official record</a> | The title and publisher scope identify CATALYST as a likely core LLM thematic-analysis system, but the Taylor & Francis full article/PDF was purchase-only and no lawful open manuscript was found. No methodological or numerical evidence is extracted from the abstract. |
| focused forward citation chaining / doi:10.1145/3711022               | LATA: A Pilot Study on LLM-Assisted Thematic Analysis of Online Social Network Data Generation Experiences | <a href="#">DOI</a>             | Sought but not retrieved after documented attempts; no full-text methodological extraction or quality score was created.                                                                                                                                                    |
| focused forward citation chaining / doi:10.1007/978-3-031-60533-8_14  | Future Directions in Qualitative Research                                                                  | <a href="#">DOI</a>             | Sought but not retrieved after documented attempts; no full-text methodological extraction or quality score was created.                                                                                                                                                    |
| focused forward citation chaining / doi:10.1007/978-3-031-65735-1_19  | From Words to Themes: AI-Powered Qualitative Data Coding and Analysis                                      | <a href="#">DOI</a>             | Sought but not retrieved after documented attempts; no full-text methodological extraction or quality score was created.                                                                                                                                                    |
| focused forward citation chaining / doi:10.1386/i_scc_00058_1         | Computer-assisted qualitative visual analysis: Automating thematic analysis of images                      | <a href="#">DOI</a>             | Sought but not retrieved after documented attempts; no full-text methodological extraction or quality score was created.                                                                                                                                                    |
| backward citation chaining / laato_2025_gioia_cot                     | AMCIS 2025 paper 2130; AISel article 1249; no DOI reported                                                 | not reported                    | Likely core from official abstract/metadata, but protocol forbids methodological extraction or inclusion without complete full-text inspection.                                                                                                                             |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

### Publication/version-family resolutions (9 unique identities)

These records were resolved into a preferred family member or existing assessed identity and were not counted as additional evidence rows.

| Route/ID                                          | Candidate title                                                                                                          | Source              | Resolution                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-0016</a> | A Multi-Modular Human-AI Workflow for LLM-Assisted Thematic Analysis: Application to COPD Telerehabilitation Interviews. | <a href="#">DOI</a> | This short COPD component paper is subsumed by the later same-team comparative evaluation across ILD, POTS, and COPD, already assessed in master evidence as bai_2026_three_datasets (DOI 10.2196/96129). No independent evidence or quality row is created.                                                                                                                        |
| database/register union / <a href="#">TA-0336</a> | Narrative-Integrated Thematic Analysis (NITA): How can LLMs support theme generation without coding?                     | <a href="#">DOI</a> | This inspected OSF preprint belongs to the same NITA publication family as the fuller peer-reviewed 2026 Qualitative Research in Psychology version of record extracted in research/screening_2025_fulltext_extractions.json as nguyen_trung_nguyen_2026_nita (DOI 10.1080/14780887.2026.2638348). Retain the VOR extraction once and suppress this duplicate evidence/quality row. |
| database/register union / <a href="#">TA-0477</a> | Workflow Strategy in a Modular Human-AI Collaboration Framework for LLM-Assisted Thematic Analysis of POTS Interviews    | <a href="#">DOI</a> | This POTS component paper is subsumed by the later same-team ILD/POTS/COPD comparative evaluation already assessed in master evidence as bai_2026_three_datasets (DOI 10.2196/96129). No independent evidence or quality row is created.                                                                                                                                            |
| database/register union / <a href="#">TA-1113</a> | QualiGPT: GPT as an easy-to-use tool for qualitative coding                                                              | not reported        | This QualiGPT arXiv paper is an earlier version/tool-family record of the later 2024 QualiGPT study at TA-1049; it is not counted as a separate independent study without a dependence audit.                                                                                                                                                                                       |

| Route/ID                                          | Candidate title                                                                                                                          | Source       | Resolution                                                                                                                                                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| database/register union / <a href="#">TA-1271</a> | Agent-as-Peer-Debriefing: A Multi-Agent Framework with Perspective-Based Refinement for Qualitative Analysis                             | not reported | Agent-as-Peer-Debriefing and TA-1319 LLM-as-Peer-Debriefing report the same three-dataset, three-model multi-perspective framework; TA-1319 is retained as the preferred current title pending full-text/version verification. |
| database/register union / <a href="#">TA-1273</a> | "Always Nice and Confident, Sometimes Wrong": Developer's Experiences Engaging Generative AI Chatbots Versus Human-Powered Q&A Platforms | not reported | This title and abstract duplicate TA-1060 ('Always Nice and Confident, Sometimes Wrong'); both study developers' experiences with LLM tools rather than automated qualitative analysis.                                        |
| database/register union / <a href="#">TA-1285</a> | CentaurTA Studio: A Self-Improving Human-Agent Collaboration System for Thematic Analysis                                                | not reported | CentaurTA is already represented and quality-scored in master evidence as centaurta_2026; no duplicate evidence or quality row is created.                                                                                     |
| database/register union / <a href="#">TA-1299</a> | Exploring the Potential of Human-LLM Synergy in Advancing Qualitative Analysis: A Case Study on Mental-Illness Stigma                    | not reported | Same CHALET dependency/version family as TA-0955; the TOCHI article is retained once as the preferred record and this earlier title/version receives no separate evidence or quality row.                                      |
| database/register union / <a href="#">TA-1316</a> | Landscape of Generative AI in Global News: Topics, Sentiments, and Spatiotemporal Analysis                                               | not reported | The complete arXiv text, authors, 24,827-article dataset, methods, and results match TA-1317; TA-1317 is the ICWSM 2024 proceedings version and is the preferred family record.                                                |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

### 19.3 Complete evidence-extraction table

The frozen extraction is [research/master\\_evidence.json](#): **237 rows × exactly 52 fields**, one row per retained publication identity. Adjacent/contextual records retain the same extraction schema; missing information is explicitly **not reported**. The complete field order is:

Paper ID; Full citation; Title; Authors; Year; Venue; Publication type; Peer-reviewed status; Publication history notes; URL; DOI; Citation count, source,

and retrieval date; Core or adjacent classification; Relationship/dependence family; Research objective; Claimed qualitative methodology; Actual operationalized methodology; Inductive, deductive, or hybrid; Domain; Dataset name; Public or restricted data; Number of documents; Number of participants; Approximate corpus size; Document length; Language; Sensitive-data status; Model family; Exact model/version; Prompting strategy; Fine-tuning or adaptation; Single-agent or multi-agent; Agent roles; Human role; Unit of analysis; Generated outputs; Evidence/provenance mechanism; Baselines; Number of runs or seeds; Automatic metrics; Human-evaluation design; Number and expertise of evaluators; Inter-rater reliability; Statistical analysis; Principal results; Efficiency or cost results; Ethics and privacy treatment; Reproducibility resources; Author-reported limitations; Additional validity concerns; Relationship to prior papers; Relevance to the proposed ARR project.

The master also preserves direct official URLs, DOI/status history, dataset and human-reference dependencies, and evidence-location-grounded reviewer concerns. Section 18 provides a human-readable bibliographic view of all 237 IDs; the JSON is the authoritative row-level table.

#### 19.4 Core-paper quality rubric and audit

Exactly **181 quality rows** correspond one-to-one with the 181 core evidence IDs. No noncore record is scored. Each dimension contains both a 0/1/2 score and a paper-specific rationale. Scores are interpreted against the paper's stated claim: 0=absent/inadequate, 1=partial/unclear, 2=adequate. Totals are navigation aids and must not be used to declare a universally "best" study.

| Dimension                                  | What the reviewer assessed                                                                                                    |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Clarity of its qualitative methodology     | Whether the claimed tradition, analytic contract, unit, and operational steps are explicit and methodologically congruent.    |
| Appropriateness of its dataset             | Whether the data, sampling, size, context, language, and sensitivity can support the stated claim.                            |
| Transparency of its computational workflow | Whether model/version, prompts, chunking/context, adaptation, agents, retries, and intervention points are reproducible.      |
| Strength and independence of its baselines | Whether comparators are strong, independent, leakage-safe, and evaluated on equivalent units/tasks.                           |
| Adequacy of its human evaluation           | Whether evaluator expertise/count, independence, blinding, sampling, rubric, disagreement, and uncertainty support the claim. |

| Dimension                                               | What the reviewer assessed                                                                                                                 |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Validity of its automatic metrics                       | Whether the metric operationalizes the intended construct, is human-validated, and reports units, thresholds, uncertainty, and statistics. |
| Treatment of interpretive plurality                     | Whether alternative readings, positionality, disagreement, and limits of a single reference are acknowledged or preserved.                 |
| Evidence grounding and traceability                     | Whether source-to-output links preserve exact text, speaker/context, versions, warrants, and auditable omissions.                          |
| Stability and reproducibility testing                   | Whether repeated runs or crossed perturbations test substantive—not merely lexical—stability and resources permit rerun.                   |
| Ethics, privacy, and responsible-research reporting     | Whether consent, ethics, deployment, retention/training, privacy, representation, stakeholder authority, and accountability are addressed. |
| Availability of data, prompts, code, and model versions | Whether usable data or fixtures, prompts/configuration, code, and exact model/version information are available.                           |
| Whether conclusions are proportionate to the evidence   | Whether claims remain within the evaluated task, data, method, status, denominators, and acknowledged limitations.                         |

Source: reviewer synthesis from the cited primary studies; full row-level extraction is in the companion workbook.

The claimed-versus-operationalized audit in Section 4.1 covers all 181 core records: 169 have both substantive fields, ten have neither, one is claim-only, and one operation-only. Under the stated strict lower-bound rubric, 34 receive a material mismatch/narrowing flag; those flags are reviewer judgments separate from quality totals.

## 19.5 Comparative tables and dependency fields

The main report provides separate compact tables for architectures (Section 5.2), human roles (Section 6), datasets/domains (Section 7), metric constructs and human-evaluation protocols (Section 8), reproducibility and ethics (Sections 10 and 12), quality dimensions (Section 13.4), gaps (Section 14), and the structured ARR designs (Section 15). The full 52-field master supports paper-level reconstruction of every comparison. [Relationship/dependence family](#) and [Relationship to prior papers](#) preserve preprint/VOR families, reused corpora and human themes, component papers, and secondary reviews. Important one-count rules include NITA's 2025 preprint/2026 VOR, the Bai ILD/POTS/COPD family, CentaurTA, CHALET/TOCHI, QualiGPT, the AAOCA clinical family, TA-MAS/Simon, and the event/proceedings histories described in Sections 7, 16, and 18.

## 19.6 Reproducibility inventory and frozen checks

- [research/search\\_union\\_deduplicated.json](#): 1,360-record DOI/title-deduplicated database union.

- `research/review_flow.json`: terminal title/abstract decisions, retrieval outcomes, 18 full-text exclusions, 53 unique nonretrievals, nine family resolutions, chain-parent provenance, and both identity and route-event arithmetic.
- `research/master_evidence.json`: 237 evidence rows, 181 core-only quality rows, 52 evidence fields, and 12 scored dimensions.
- `research/citation_chaining.json`: backward/forward candidate identities, parent seeds, sources, overlaps, and outcomes.
- `research/screening_2020_2024_resolution.json`, `research/screening_2025_resolution.json`, and `research/screening_2026_resolution.json`: year-specific second-pass decisions and evidence locations.

Frozen SHA-256 values used for this report: master  
 c5515b1825ba4b8af55ab665057361f0328730c94673b1f9b95e53bf3bf70ca8; review flow  
 f1f16545496ad13f2cd86f23adb423452e5c3d6770853355ab943a0149fd8483; final  
 integrity audit  
 6802e89f2981f76dd54012e6e1ea45d9426226f14a275719a034ecefdda7c741. The  
 publisher's final integrity run reported identical repeated builds and **39/39 checks passing**, including full-  
 text-exclusion reasons, nonretrieval statuses, and the Kempny evidence-map boundary.